Cisco Nexus 9336C-FX2共享交換器

ONTAP Systems Switches

NetApp
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Cisco 9336C-FX2共享交換器總覽

從S299.1開始ONTAP 、您可以使用Cisco Nexus 9336C-FX2交換器、將儲存設備和叢集功能結合到共享交換器情境中。

Cisco共享交換器的設定與組態指南

支援的交換器ONTAP

從S299.1開始ONTAP 、您可以使用Cisco Nexus 9336C-FX2交換器、將儲存設備和叢集功能結合到共享交換器組態中。如果您想要建置ONTAP 具有兩個以上節點的叢集、您需要兩個支援的網路交換器。

支援下列Cisco共用網路交換器。

Nexus 9336C-FX2

您可以將Cisco Nexus 9336C-FX2交換器（X190200/X190210）安裝在NetApp系統機櫃或第三方機櫃中、並附有交換器隨附的標準支架。

下表列出9336C-FX2交換器、風扇和電源供應器的零件編號和說明：

產品編號	說明
X190200-CS-PE	N9K-9336C-FX2、CS、PTSX、36PT10/25/40/100GQSFP28
X190200-CS-PI	N9K-9336C-FX2、CS、PSIN、36PT10/25/40/100GQSFP28
X190002	配件套件X190001/X190003
X-NXA-PAC-1100 W-PE2	N9K-9336C AC 1100 W PSU -連接埠側邊排氣氣流
X-NXA-PAC-1100 W-Pi2	N9K-9336C AC 1100 W PSU -連接埠側進氣
X-NXA-FAN-65CFM-PE	N9K-9336C 65CFM、連接埠側邊排氣氣流
X-NXA-FAN-65CFM-PI	N9K-9336C 65CFM、連接埠側進氣氣流

設定交換器

如果您還沒有必要的組態資訊和文件、您必須先收集這些資訊、然後再設定共用交換器。

開始之前

- 您必須能在安裝站台存取HTTP、FTP或TFTP伺服器、才能下載適用的NX-OS和參考組態檔（RCF）版本。
- 您必須擁有所需的共享交換器文件。

  請參閱 [Required documentation for shared switches] 以取得更多資訊。

- 您必須備有必要的控制器文件和ONTAP 參考文件。

  請參閱 "NetApp ONTAP 產品文件"。

- 您必須擁有適用的授權、網路和組態資訊、以及纜線。
- 您必須備妥完整的佈線工作表。

> 除了佈線圖形之外、本指南也提供範例工作表、其中包含建議的連接埠指派和空白工作表、可供您用來設定網路。如需詳細資訊、請參閱 "Hardware Universe"。

所有Cisco共享交換器都會以標準Cisco原廠預設組態送達。這些交換器也有NX-OS軟體的最新版本、但未載入RCFs。

> 您必須從下載適用的NetApp RCFs "NetApp 支援網站" 適用於您收到的交換器。

程序

1. 裝入交換器、控制器和NS224 NVMe儲存櫃。請參閱https://docs.netapp.com/platstor/topic/com.netapp.doc.hw-sw-9336c-install-cabinet/GUID-92287262-E7A6-4A62-B159-7F148097B33B.html["在NetApp機櫃中安裝Cisco Nexus 9336C-FX2叢集交換器和傳遞面板"] 指南、以取得在NetApp機櫃中安裝交換器的指示。
2. 開啟交換器、控制器和NS224 NVMe儲存櫃的電源。
3. 根據中提供的資訊、執行交換器的初始組態 [Required configuration information]。
4. 確認您在設定結束時所顯示的顯示器上所做的組態選擇、並確定您已儲存組態。
5. 檢查交換器上的軟體版本、必要時將NetApp支援的軟體版本下載到交換器。

   如果您下載NetApp支援的軟體版本、則必須下載NetApp網路交換器參考組態檔案、並將其與您儲存的組態合併 步驟3.。您可以從下載檔案和指示 "Cisco乙太網路交換器" 頁面。如果您有自己的交換器、請參閱 "Cisco" 網站。

必要的組態資訊

對於組態設定、您需要適當數量和類型的纜線和纜線連接器、以供交換器使用。視您初始設定的交換器類型而定、您需要使用隨附的主控台纜線連接至交換器主控台連接埠；您也需要提供特定的網路資訊。

所有交換器的必要網路資訊

- 所有交換器組態都需要下列網路資訊：
  - 用於管理網路流量的IP子網路
  - 每個儲存系統控制器和所有適用交換器的主機名稱和IP位址
  - 大部分的儲存系統控制器都是透過e0M介面來管理、方法是連接至乙太網路服務連接埠（扳手圖示）。在E0M介面上AFF 、E0M AFF 介面使用專用的乙太網路連接埠、可在ESIA800和ESIEA700s系統上使用。
- 請參閱 "Hardware Universe" 以取得最新資訊。

Cisco Nexus 9336C-FX2交換器的必要網路資訊

對於Cisco Nexus 9336C-FX2交換器、您必須在第一次啟動交換器時、針對下列初始設定問題提供適當的回應。您站台的安全性原則定義了回應和服務、以便：

1. 中止自動資源配置並繼續正常設定？（是/否）

   回應* yes *。預設值為「否」

2. 是否要強制執行安全密碼標準？（是/否）

   回應* yes *。預設值為yes。

3. 輸入admin的密碼。

   預設密碼為admin；您必須建立新的強式密碼。

   弱密碼可能會遭到拒絕。

4. 是否要進入基本組態對話方塊？（是/否）

   在交換器的初始組態中回應* yes *。

5. 建立另一個登入帳戶？（是/否）

   您的答案取決於您站台的原則、取決於替代系統管理員。預設值為「否」

6. 設定唯讀SNMP社群字串？（是/否）

   回應*否*。預設值為「否」

7. 設定讀寫SNMP社群字串？（是/否）

   回應*否*。預設值為「否」

8. 輸入交換器名稱。

   交換器名稱上限為63個英數字元。

9. 是否繼續頻外（mgmt0）管理組態？（是/否）

   在該提示字元中以* yes *（預設值）回應。在mgmt0 ipv4位址：提示字元中、輸入您的IP位址：ip_address

10. 設定預設閘道？（是/否）

    回應* yes *。在「Default-gateway:（預設閘道：）」提示字元的IPV4位址、輸入您的預設閘道。

11. 設定進階IP選項？（是/否）

    回應*否*。預設值為「否」

12. 啟用Telnet服務？（是/否）

    回應*否*。預設值為「否」

13. 啟用SSH服務？（是/否）

    回應* yes *。預設值為yes。

> 建議在使用叢集交換器健全狀況監視器（CSHM）進行記錄收集功能時使用SSH。我們也建議使用SSHv2來增強安全性。

1. [[step14]輸入您要產生的SSH金鑰類型（DSA/RSA/rsa1）。預設值為RSA。
2. 輸入金鑰位元數（1024-2048）。
3. 設定NTP伺服器？（是/否）

   回應*否*。預設值為「否」

4. 設定預設介面層（L3/L2）：

   回應* L2*。預設值為L2。

5. 設定預設交換器連接埠介面狀態（關機/節點關機）：

   使用* noshut*回應。預設值為noshut。

6. 設定CoPP系統設定檔（嚴格/中等/輕度/高密度）：

   回應*嚴格*。預設為嚴格。

7. 是否要編輯組態？（是/否）

   此時您應該會看到新的組態。檢閱您剛輸入的組態、並進行必要的變更。如果您對組態感到滿意、請在提示時回答「否」。如果您要編輯組態設定、請使用* yes *回應。

8. 使用此組態並加以儲存？（是/否）

   回應* yes *以儲存組態。這會自動更新Kickstart和系統映像。

   > 如果您在此階段未儲存組態、下次重新啟動交換器時、將不會有任何變更生效。

如需交換器初始組態的詳細資訊、請參閱下列指南："Cisco Nexus 9336C-FX2安裝與升級指南"。

共享交換器所需的文件

您需要特定的交換器和控制器文件來設定ONTAP 您的整套功能。

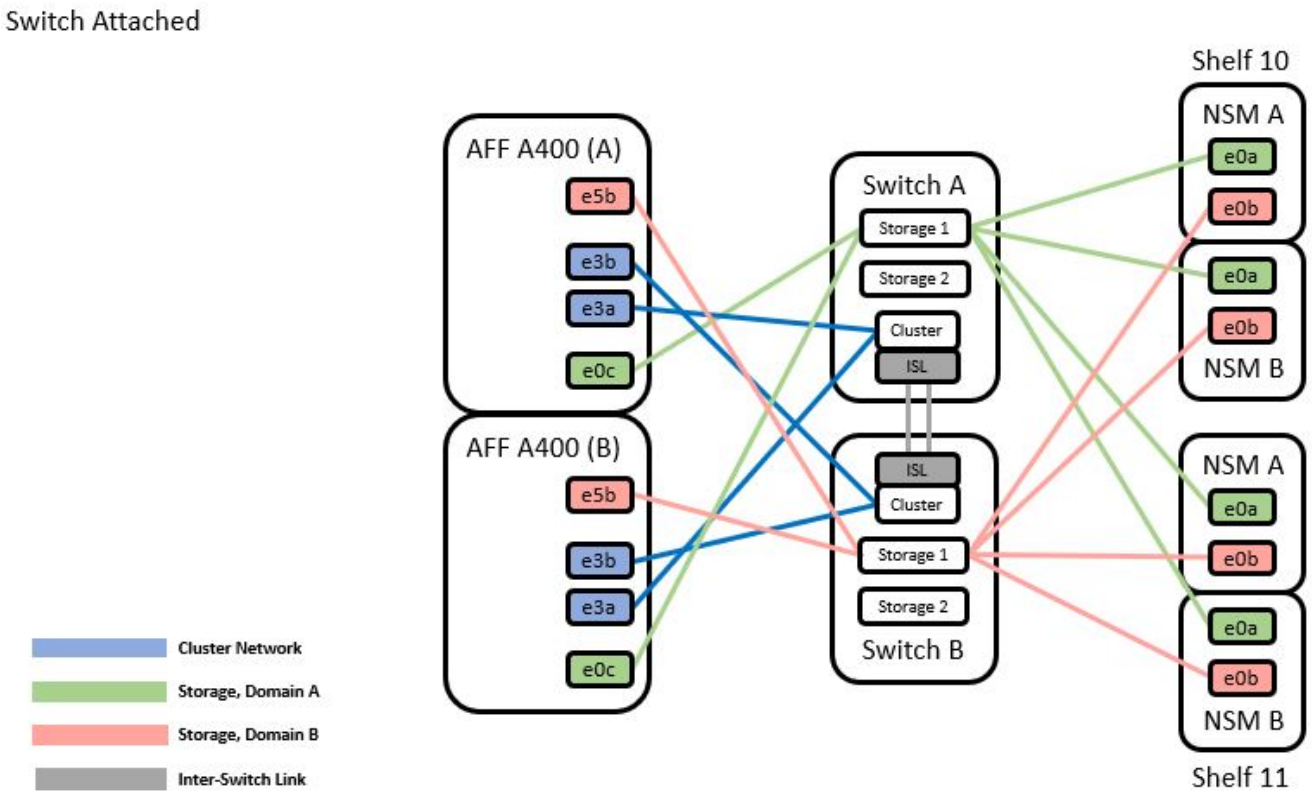
若要設定Cisco Nexus 9336C-FX2共用交換器、請參閱 "Cisco Nexus 9000系列交換器支援" 頁面。

文件標題	說明
"Nexus 9000系列硬體安裝指南"	提供有關站台需求、交換器硬體詳細資料及安裝選項的詳細資訊。
"Cisco Nexus 9000系列交換器軟體組態指南"（請選擇安裝在交換器上的NX-OS版本指南）	提供您需要的初始交換器組態資訊、然後才能設定交換器ONTAP 以供執行故障操作。
"Cisco Nexus 9000系列NX-OS軟體升級與降級指南"（請選擇安裝在交換器上的NX-OS版本指南）	如ONTAP 有必要、提供如何將交換器降級至支援的交換器軟體的相關資訊。

文件標題	說明
"Cisco Nexus 9000系列NX-OS命令參考主索引"	提供Cisco所提供之各種命令參考資料的連結。
"Cisco Nexus 9000 MIBs參考資料"	說明Nexus 9000交換器的管理資訊庫（MIB）檔案。
"Nexus 9000系列NX-OS系統訊息參考"	說明Cisco Nexus 9000系列交換器的系統訊息、資訊訊息、以及其他可能有助於診斷連結、內部硬體或系統軟體問題的訊息。
"Cisco Nexus 9000系列NX-OS版本資訊" （請針對安裝在交換器上的NX-OS版本選擇附註）	說明Cisco Nexus 9000系列的功能、錯誤和限制。
"Cisco Nexus 9000系列的法規遵循與安全資訊"	提供Nexus 9000系列交換器的國際機構法規遵循、安全及法規資訊。

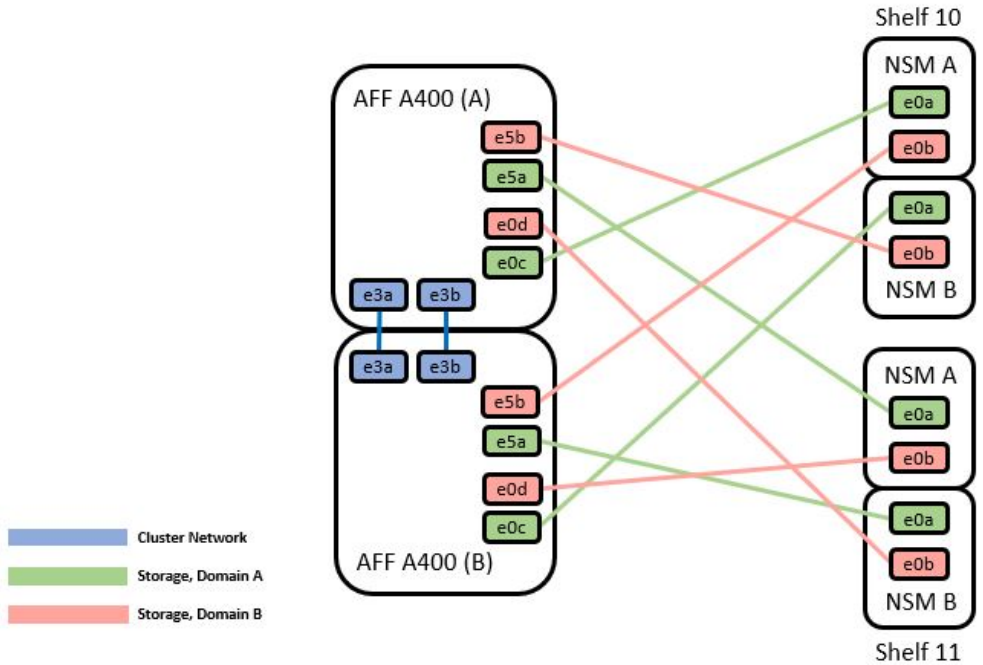
Cisco Nexus 9336C-FX2纜線詳細資料

您可以使用下列纜線映像來完成控制器與交換器之間的纜線連接。如果您想要將NS224儲存設備連接成交換器、請依照交換器附加的圖表進行：



如果您想要將NS224儲存設備連接成直接附加的連接埠、而非使用共用交換器儲存連接埠、請依照直接附加的圖表進行：

Direct Attached



	Port	Node
Path_A	e0c	Node_A
		Node_B
	e5a	Node_A
		Node_B
Path_B	e5b	Node_A
		Node_B
	e0d	Node_A
		Node_B

Cisco Nexus 9336C-FX2纜線工作表

如果您想要記錄支援的平台、您必須使用完整的佈線工作表範例作為指南來填寫空白的佈線工作表。

每對交換器的連接埠定義範例如下
：

	Switch A			Switch B	
Switch Port	Port Role	Port Usage	Switch Port	Port Role	Port Usage
1	Cluster	40/100GbE	1	Cluster	40/100GbE
2	Cluster	40/100GbE	2	Cluster	40/100GbE
3	Cluster	40/100GbE	3	Cluster	40/100GbE
4	Cluster	40/100GbE	4	Cluster	40/100GbE
5	Cluster	40/100GbE	5	Cluster	40/100GbE
6	Cluster	40/100GbE	6	Cluster	40/100GbE
7	Cluster	40/100GbE	7	Cluster	40/100GbE
8	Cluster	40/100GbE	8	Cluster	40/100GbE
9	Cluster	40GbE w/4x10GbE b/o	9	Cluster	40GbE w/4x10GbE b/o
10	Cluster	100GbE w/4x25GbE b/o	10	Cluster	100GbE w/4x25GbE b/o
11	Storage	100GbE	11	Storage	100GbE
12	Storage	100GbE	12	Storage	100GbE
13	Storage	100GbE	13	Storage	100GbE
14	Storage	100GbE	14	Storage	100GbE
15	Storage	100GbE	15	Storage	100GbE
16	Storage	100GbE	16	Storage	100GbE
17	Storage	100GbE	17	Storage	100GbE
18	Storage	100GbE	18	Storage	100GbE
19	Storage	100GbE	19	Storage	100GbE
20	Storage	100GbE	20	Storage	100GbE
21	Storage	100GbE	21	Storage	100GbE
22	Storage	100GbE	22	Storage	100GbE
23	Storage	100GbE	23	Storage	100GbE
24	Storage	100GbE	24	Storage	100GbE
25	Storage	100GbE	25	Storage	100GbE
26	Storage	100GbE	26	Storage	100GbE
27	Storage	100GbE	27	Storage	100GbE
28	Storage	100GbE	28	Storage	100GbE
29	Storage	100GbE	29	Storage	100GbE
30	Storage	100GbE	30	Storage	100GbE
31	Storage	100GbE	31	Storage	100GbE
32	Storage	100GbE	32	Storage	100GbE
33	Storage	100GbE	33	Storage	100GbE
34	Storage	100GbE	34	Storage	100GbE
35	ISL	100GbE	35	ISL	100GbE
36	ISL	100GbE	36	ISL	100GbE

其中：

- 100G ISL至交換器A連接埠35
- 100G ISL至交換器A連接埠36
- 100G ISL至交換器B連接埠35
- 100G ISL至交換器B連接埠36

空白的佈線工作表

您可以使用空白的佈線工作表來記錄叢集中支援作為節點的平台。支援的叢集連線表Hardware Universe 定義平台所使用的叢集連接埠。

	Switch A			Switch B	
Switch Port	Port Role	Port Usage	Switch Port	Port Role	Port Usage
1			1		
2			2		
3			3		
4			4		
5			5		
6			6		
7			7		
8			8		
9			9		
10			10		
11			11		
12			12		
13			13		
14			14		
15			15		
16			16		
17			17		
18			18		
19			19		
20			20		
21			21		
22			22		
23			23		
24			24		
25			25		
26			26		
27			27		
28			28		
29			29		
30			30		
31			31		
32			32		
33			33		
34			34		
35			35		
36			36		

其中：

- 100G ISL至交換器A連接埠35
- 100G ISL至交換器A連接埠36
- 100G ISL至交換器B連接埠35
- 100G ISL至交換器B連接埠36

安裝NX-OS軟體和RCT

在Cisco Nexus 9336C-FX2交換器上安裝NX-OS軟體和RCF

Cisco NX-OS軟體與參考組態檔（RCF）必須安裝在Cisco Nexus 9336C-FX2叢集交換器上。

在叢集交換器上安裝NX-OS軟體和RCF之前、必須符合下列條件：

- 叢集必須完全正常運作（記錄中不應有錯誤或類似問題）。
- 如果您只安裝NX-OS並保留目前的RCF版本、則必須在RCF中檢查或設定所需的開機組態、以反映所需的開機映像。
- 如果您需要變更開機組態以反映目前的開機映像、則必須在重新套用RCF之前執行此動作、以便在未來重新開機時產生正確的版本。
- 安裝RCF時、您必須有與交換器的主控台連線。
- 您必須查閱Cisco乙太網路交換器頁面上的交換器相容性表、以取得支援ONTAP 的版本、包括支援的版本、作業系統和RCF。請參閱 "Cisco乙太網路交換器" 以取得更多資訊。
- RCF中的命令語法與NX-OS版本中的命令語法之間可能存在相依性。
- 您必須參閱Cisco網站上提供的適當軟體與升級指南、以取得Cisco Nexus 9000系列交換器上Cisco交換器升級與降級程序的完整文件。請參閱 "Cisco Nexus 9000系列交換器" 以取得更多資訊。
- 您必須擁有目前的RCF。

本程序中的範例使用兩個節點。這些節點使用兩個100GbE叢集互連連接埠e3a和e3b、如同A400控制器。請參閱 "Hardware Universe" 驗證平台上的叢集連接埠是否正確。

> 命令輸出可能會因ONTAP 不同版本的不相同而有所差異。

本程序中的範例使用下列交換器和節點命名法：

- 這兩個Cisco交換器的名稱為_CS1_和_CS2_。
- 節點名稱為_cluster1-01_和_cluster1-02_。
- 叢集LIF的名稱為_cluster1-01_clus1_、而叢集1-01_clus2_則為叢集1-01和叢集1-02_clus1_、叢集1-02_clus2_則為叢集1-02。
- cluster1：：*>提示會指出叢集的名稱。

> 此程序需要同時使用ONTAP 支援指令和Cisco Nexus 9000系列交換器的指令；ONTAP 除非另有說明、否則會使用支援指令。

步驟

1. 如果AutoSupport 此叢集啟用了Section,則會叫用AutoSupport 下列訊息來抑制自動案例建立:「System nodeeseInvoke AutoSupport -Node *-type all -most mn=x h」

   其中x是維護時段的持續時間（以小時為單位）。

2. 將權限層級變更為進階、並在系統提示您繼續時輸入y：「et -priv榮幸advanced'（設定-權限進階）」

   出現進階提示（*>）。

3. 顯示每個叢集互連交換器的每個節點已設定多少個叢集互連介面：「network device-dDiscovery show

-protocol cup」

```
cluster1:: network device-discovery show -protocol cdp
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------- --------
cluster1-02/cdp
            e3a    cs1                       Eth1/2            N9K-C9336C
            e3b    cs2                       Eth1/2            N9K-C9336C
cluster1-01/cdp
            e3a    cs1                       Eth1/1            N9K-C9336C
            e3b    cs2                       Eth1/1            N9K-C9336C
4 entries were displayed.
```

1. 檢查每個叢集介面的管理或作業狀態：
   a. 顯示網路連接埠屬性：「network port show–IPSpace Cluster（網路連接埠顯示–IPSpace叢集）」

```
cluster1::*> network port show -ipspace Cluster
Node: cluster1-02
                                                   Speed(Mbps)  Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
--------- ------------ ---------------- ---- ---- ------------ ------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
e3b       Cluster      Cluster          up   9000  auto/100000 healthy

Node: cluster1-01
                                                   Speed(Mbps)  Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
--------- ------------ ---------------- ---- ---- ------------ ------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
e3b       Cluster      Cluster          up   9000  auto/100000 healthy
4 entries were displayed.
```

1. 顯示有關生命的資訊：「網路介面show - vserver叢集」

```
cluster1::*> network interface show -vserver Cluster
        Logical            Status     Network            Current
Current Is
Vserver Interface          Admin/Oper Address/Mask       Node        Port
Home
------- ------------------ ---------- ------------------ -----------
------ ----
Cluster
        cluster1-01_clus1  up/up      169.254.209.69/16  cluster1-01  e3a
true
        cluster1-01_clus2  up/up      169.254.49.125/16  cluster1-01  e3b
true
        cluster1-02_clus1  up/up      169.254.47.194/16  cluster1-02  e3a
true
        cluster1-02_clus2  up/up      169.254.19.183/16  cluster1-02  e3b
true
4 entries were displayed.
```

1. Ping遠端叢集lifs：「叢集ping叢集節點節點名稱」

```
cluster1::*> cluster ping-cluster -node cluster1-02
Host is cluster1-02
Getting addresses from network interface table...
Cluster cluster1-01_clus1 169.254.209.69 cluster1-01     e3a
Cluster cluster1-01_clus2 169.254.49.125 cluster1-01     e3b
Cluster cluster1-02_clus1 169.254.47.194 cluster1-02     e3a
Cluster cluster1-02_clus2 169.254.19.183 cluster1-02     e3b
Local = 169.254.47.194 169.254.19.183
Remote = 169.254.209.69 169.254.49.125
Cluster Vserver Id = 4294967293
Ping status:
....
Basic connectivity succeeds on 4 path(s)
Basic connectivity fails on 0 path(s)
................
Detected 9000 byte MTU on 4 path(s):
    Local 169.254.19.183 to Remote 169.254.209.69
    Local 169.254.19.183 to Remote 169.254.49.125
    Local 169.254.47.194 to Remote 169.254.209.69
    Local 169.254.47.194 to Remote 169.254.49.125
Larger than PMTU communication succeeds on 4 path(s)
RPC status:
2 paths up, 0 paths down (tcp check)
2 paths up, 0 paths down (udp check)
```

1. 確認所有叢集lifs上都已啟用自動還原命令：「network interface show - vserver cluster -Fields autosf還原」（網路介面show - vserver叢集-Fields自動還原）

```
cluster1::*> network interface show -vserver Cluster -fields auto-revert
          Logical
Vserver   Interface            Auto-revert
--------- -------------------- ------------
Cluster
          cluster1-01_clus1    true
          cluster1-01_clus2    true
          cluster1-02_clus1    true
          cluster1-02_clus2    true
4 entries were displayed.
```

1. 使用下列命令、啟用乙太網路交換器健全狀況監視器記錄收集功能、以收集交換器相關的記錄檔：
   - 「系統交換器乙太網路記錄設定密碼」
   - 「系統交換器乙太網路記錄啟用收集」

```
cluster1::*> system switch ethernet log setup password
Enter the switch name: <return>
The switch name entered is not recognized.
Choose from the following list:
cs1
cs2
cluster1::*> system switch ethernet log setup-password
Enter the switch name: cs1
RSA key fingerprint is e5:8b:c6:dc:e2:18:18:09:36:63:d9:63:dd:03:d9:cc
Do you want to continue? {y|n}::[n] y
Enter the password: <enter switch password>
Enter the password again: <enter switch password>
cluster1::*> system switch ethernet log setup-password
Enter the switch name: cs2
RSA key fingerprint is 57:49:86:a1:b9:80:6a:61:9a:86:8e:3c:e3:b7:1f:b1
Do you want to continue? {y|n}:: [n] y
Enter the password: <enter switch password>
Enter the password again: <enter switch password>
cluster1::*> system  switch ethernet log enable-collection
Do you want to enable cluster log collection for all nodes in the cluster?
{y|n}: [n] y
Enabling cluster switch log collection.
cluster1::*>
```

如果這些命令中有任何一個出現錯誤、請聯絡NetApp支援部門。

在Cisco Nexus 9336C - FX2叢集交換器上安裝NX-OS軟體

您可以使用此程序、在Cisco Nexus 9336C-FX2叢集交換器上安裝NX-OS軟體。

步驟

1. 將叢集交換器連線至管理網路。
2. 使用「ping」命令來驗證與裝載NX-OS軟體和RCF之伺服器的連線。

   此範例可驗證交換器是否能以IP位址「172.19.2.1（IP位址：

```
cs2# ping 172.19.2.1
Pinging 172.19.2.1 with 0 bytes of data:
Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
```

1. [[step3]將NX-OS軟體和EPLD映像複製到Nexus 9336C-FX2交換器。

```
cs2# copy sftp: bootflash: vrf management
Enter source filename: /code/nxos.9.3.5.bin
Enter hostname for the sftp server: 172.19.2.1
Enter username: user1
Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/nxos.9.3.5.bin  /bootflash/nxos.9.3.5.bin
/code/nxos.9.3.5.bin  100% 1261MB   9.3MB/s   02:15
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
cs2# copy sftp: bootflash: vrf management
Enter source filename: /code/n9000-epld.9.3.5.img
Enter hostname for the sftp server: 172.19.2.1
Enter username: user1
Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/n9000-epld.9.3.5.img  /bootflash/n9000-epld.9.3.5.img
/code/n9000-epld.9.3.5.img  100%  161MB   9.5MB/s   00:16
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
```

1. [[step4]驗證NX-OS軟體的執行版本：「How version」（顯示版本）

```
cs2# show version
Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
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unless
otherwise stated, there is no warranty, express or implied, including but
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limited to warranties of merchantability and fitness for a particular
purpose.
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GNU General Public License (GPL) version 3.0  or the GNU
Lesser General Public License (LGPL) Version 2.1 or
Lesser General Public License (LGPL) Version 2.0.
A copy of each such license is available at
http://www.opensource.org/licenses/gpl-2.0.php and
http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.
Software
  BIOS: version 08.38
  NXOS: version 9.3(4)
  BIOS compile time:  05/29/2020
  NXOS image file is: bootflash:///nxos.9.3.4.bin
  NXOS compile time:  4/28/2020 21:00:00 [04/29/2020 02:28:31]
Hardware
  cisco Nexus9000 C9336C-FX2 Chassis
  Intel(R) Xeon(R) CPU E5-2403 v2 @ 1.80GHz with 8154432 kB of memory.
  Processor Board ID FOC20291J6K
  Device name: cs2
  bootflash:   53298520 kB
Kernel uptime is 0 day(s), 0 hour(s), 3 minute(s), 42 second(s)
Last reset at 157524 usecs after Mon Nov  2 18:32:06 2020
  Reason: Reset Requested by CLI command reload
  System version: 9.3(4)
  Service:
plugin
  Core Plugin, Ethernet Plugin
Active Package(s):

cs2#
```

1. [[step5]安裝NX-OS映像。

> 安裝映像檔會在每次重新開機時載入映像檔。

```
cs2# install all nxos bootflash:nxos.9.3.5.bin
Installer will perform compatibility check first. Please wait.
Installer is forced disruptive
Verifying image bootflash:/nxos.9.3.5.bin for boot variable "nxos".
[####################] 100% -- SUCCESS
Verifying image type.
[####################] 100% -- SUCCESS
Preparing "nxos" version info using image bootflash:/nxos.9.3.5.bin.
[####################] 100% -- SUCCESS
Preparing "bios" version info using image bootflash:/nxos.9.3.5.bin.
[####################] 100% -- SUCCESS
Performing module support checks.
[####################] 100% -- SUCCESS
Notifying services about system upgrade.
[####################] 100% -- SUCCESS
Compatibility check is done:
Module  bootable       Impact     Install-type  Reason
------  --------  --------------- ------------  ------
  1       yes      disruptive         reset     default upgrade is not
hitless
Images will be upgraded according to following table:
Module   Image    Running-Version(pri:alt                New-Version
Upg-

Required
------- --------- ---------------------------------------
------------------- ------------
  1      nxos     9.3(4)                                  9.3(5)
yes
  1      bios     v08.37(01/28/2020):v08.23(09/23/2015)
v08.38(05/29/2020)  yes
Switch will be reloaded for disruptive upgrade.
Do you want to continue with the installation (y/n)?  [n] y
Install is in progress, please wait.
Performing runtime checks.
[####################] 100% -- SUCCESS
Setting boot variables.
[####################] 100% -- SUCCESS
Performing configuration copy.
[####################] 100% -- SUCCESS
Module 1: Refreshing compact flash and upgrading bios/loader/bootrom.
Warning: please do not remove or power off the module at this time.
[####################] 100% -- SUCCESS
Finishing the upgrade, switch will reboot in 10 seconds.
```

1. [[step6]切換器重新開機後、請驗證新版NX-OS軟體：「How version」（顯示版本）

```
cs2# show version
Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
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Lesser General Public License (LGPL) Version 2.0.
A copy of each such license is available at
http://www.opensource.org/licenses/gpl-2.0.php and
http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.
Software
  BIOS: version 05.33
  NXOS: version 9.3(5)
  BIOS compile time:  09/08/2018
  NXOS image file is: bootflash:///nxos.9.3.5.bin
  NXOS compile time:  11/4/2018 21:00:00 [11/05/2018 06:11:06]
Hardware
  cisco Nexus9000 C9336C-FX2 Chassis
  Intel(R) Xeon(R) CPU E5-2403 v2 @ 1.80GHz with 8154432 kB of memory.
  Processor Board ID FOC20291J6K
  Device name: cs2
  bootflash:   53298520 kB
Kernel uptime is 0 day(s), 0 hour(s), 3 minute(s), 42 second(s)
Last reset at 277524 usecs after Mon Nov  2 22:45:12 2020
  Reason: Reset due to upgrade
  System version: 9.3(4)
  Service:
plugin
  Core Plugin, Ethernet Plugin
Active Package(s):
```

1. [[step7]升級EPLD映像、然後重新啟動交換器。

```
cs2# show version module 1 epld
EPLD Device                      Version
---- -------------------------- -------
MI   FPGA                        0x7
IO   FPGA                        0x17
MI   FPGA2                       0x2
GEM  FPGA                        0x2
GEM  FPGA                        0x2
GEM  FPGA                        0x2
GEM  FPGA                        0x2
cs2# install epld bootflash:n9000-epld.9.3.5.img module 1
Compatibility check:
Module      Type        Upgradable  Impact      Reason
------  -------------- ------------ ----------- -------
     1      SUP        Yes          disruptive  Module Upgradable
Retrieving EPLD versions.... Please wait.
Images will be upgraded according to following table:
Module  Type   EPLD              Running-Version   New-Version  Upg-
Required
------- ------ ----------------- ----------------- ------------
------------
     1  SUP    MI FPGA           0x07              0x07         No
     1  SUP    IO FPGA           0x17              0x19         Yes
     1  SUP    MI FPGA2          0x02              0x02         No
The above modules require upgrade.
The switch will be reloaded at the end of the upgrade
Do you want to continue (y/n) ?  [n] y
Proceeding to upgrade Modules.
Starting Module 1 EPLD Upgrade
Module 1 : IO FPGA [Programming] : 100.00% (     64 of      64 sectors)
Module 1 EPLD upgrade is successful.
Module   Type  Upgrade-Result
-------- ----- --------------
     1   SUP   Success
EPLDs upgraded.
Module 1 EPLD upgrade is successful.
```

1. [[step8]交換器重新開機後、請再次登入、並確認新版的EPLD已成功載入。

```
cs2# show version module 1 epld
EPLD Device                      Version
---- -------------------------- --------
MI   FPGA                        0x7
IO   FPGA                        0x19
MI   FPGA2                       0x2
GEM  FPGA                        0x2
GEM  FPGA                        0x2
GEM  FPGA                        0x2
GEM  FPGA                        0x2
```

將RCF安裝在Cisco Nexus 9336C - FX2叢集交換器上

您可以在初次設定Nexus 9336C-FX2叢集交換器之後安裝RCF。您也可以使用此程序來升級叢集交換器上的RCF版本。

本程序中的範例使用下列交換器和節點命名法：

- 兩個Cisco交換器的名稱分別為「CS1」和「CS2」。
- 節點名稱包括"cluster1-01"、"cluster1-02"、"cluster1-03"和"cluster1-04"。
- 叢集LIF名稱為「cluster1-01_clus1」、「cluster1-01_clus2」、「cluster1-02_clus1」、「cluster1-02_clus2」、「cluster1-03_clus1」、"cluster1-03_clus2"、"cluster1-04_clus1"和"cluster1-04_clus2"。
- cluster1：：*>提示會指出叢集的名稱。

> 此程序需要同時使用ONTAP 支援指令和Cisco Nexus 9000系列交換器的指令；ONTAP 除非另有說明、否則會使用支援指令。

步驟

1. 顯示連接至叢集交換器的每個節點上的叢集連接埠：「網路裝置探索秀」

```
cluster1::*> network device-discovery show
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------  --------
cluster1-01/cdp
            e3a    cs1                       Ethernet1/7       N9K-C9336C
            e0d    cs2                       Ethernet1/7       N9K-C9336C
cluster1-02/cdp
            e3a    cs1                       Ethernet1/8       N9K-C9336C
            e0d    cs2                       Ethernet1/8       N9K-C9336C
cluster1-03/cdp
            e3a    cs1                       Ethernet1/1/1     N9K-C9336C
            e3b    cs2                       Ethernet1/1/1     N9K-C9336C
cluster1-04/cdp
            e3a    cs1                       Ethernet1/1/2     N9K-C9336C
            e3b    cs2                       Ethernet1/1/2     N9K-C9336C
cluster1::*>
```

1. 檢查每個叢集連接埠的管理和作業狀態。
2. 確認所有叢集連接埠都處於正常狀態：「網路連接埠顯示–角色叢集」

```
cluster1::*> network port show -role cluster
Node: cluster1-01

Ignore
                                                     Speed(Mbps) Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
Status
--------- ------------ ---------------- ---- ---- ----------- --------
------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
false
e0d       Cluster      Cluster          up   9000  auto/100000 healthy
false
Node: cluster1-02

Ignore
                                                     Speed(Mbps) Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper  Status
Status
--------- ------------ ---------------- ---- ---- ----------- --------
------
```

```
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
false
e0d       Cluster      Cluster          up   9000  auto/100000 healthy
false
8 entries were displayed.
Node: cluster1-03


Ignore
                                                   Speed(Mbps)  Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
Status
--------- ------------ ---------------- ---- ---- ------------ --------
------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
false
e3b       Cluster      Cluster          up   9000  auto/100000 healthy
false
Node: cluster1-04

Ignore
                                                   Speed(Mbps)  Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
Status
--------- ------------ ---------------- ---- ---- ------------ --------
------
e0a       Cluster      Cluster          up   9000  auto/100000 healthy
false
e0b       Cluster      Cluster          up   9000  auto/100000 healthy
false
cluster1::*>
```

1. 確認所有叢集介面（lifs）都位於主連接埠：「網路介面show -role cluster」（網路介面show -role cluster）

```
cluster1::*> network interface show -role cluster
        Logical            Status     Network         Current      Current
Is
Vserver Interface          Admin/Oper Address/Mask    Node         Port
Home
------- ------------------ ---------- --------------- ------------ -------
----
Cluster
        cluster1-01_clus1  up/up     169.254.3.4/23   cluster1-01  e3a
true
        cluster1-01_clus2  up/up     169.254.3.5/23   cluster1-01  e0d
true
        cluster1-02_clus1  up/up     169.254.3.8/23   cluster1-02  e3a
true
        cluster1-02_clus2  up/up     169.254.3.9/23   cluster1-02  e0d
true
        cluster1-03_clus1  up/up     169.254.1.3/23   cluster1-03  e3a
true
        cluster1-03_clus2  up/up     169.254.1.1/23   cluster1-03  e3b
true
        cluster1-04_clus1  up/up     169.254.1.6/23   cluster1-04  e3a
true
        cluster1-04_clus2  up/up     169.254.1.7/23   cluster1-04  e3b
true
8 entries were displayed.
cluster1::*>
```

1. [[step5]驗證叢集是否顯示兩個叢集交換器的資訊：「系統叢集交換器show -is監控啟用運作的true」

```
cluster1::*> system cluster-switch show -is-monitoring-enabled-operational
true
Switch                      Type               Address          Model
--------------------------- ------------------ ---------------- -----
cs1                         cluster-network    10.233.205.90    N9K-C9336C
     Serial Number: FOCXXXXXXGD
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
    Version Source: CDP
cs2                         cluster-network    10.233.205.91    N9K-C9336C
     Serial Number: FOCXXXXXXGS
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
    Version Source: CDP
cluster1::*>
```

1. 停用叢集生命體上的自動還原功能。

```
cluster1::*> network interface modify -vserver Cluster -lif * -auto-revert
false
```

1. [[step7]在叢集交換器CS2上、關閉連接至節點叢集連接埠的連接埠。

```
cs2(config)# interface eth1/1/1-2,eth1/7-8
cs2(config-if-range)# shutdown
```

1. 確認叢集LIF已移轉至叢集交換器CS1上裝載的連接埠。這可能需要幾秒鐘的時間：「網路介面show -role cluster」

```
cluster1::*> network interface show -role cluster
        Logical           Status     Network          Current      Current
Is
Vserver Interface         Admin/Oper Address/Mask     Node         Port
Home
------- ----------------- --------- ---------------- ------------- -------
----
Cluster
        cluster1-01_clus1 up/up     169.254.3.4/23   cluster1-01   e3a
true
        cluster1-01_clus2 up/up     169.254.3.5/23   cluster1-01   e3a
false
        cluster1-02_clus1 up/up     169.254.3.8/23   cluster1-02   e3a
true
        cluster1-02_clus2 up/up     169.254.3.9/23   cluster1-02   e3a
false
        cluster1-03_clus1 up/up     169.254.1.3/23   cluster1-03   e3a
true
        cluster1-03_clus2 up/up     169.254.1.1/23   cluster1-03   e3a
false
        cluster1-04_clus1 up/up     169.254.1.6/23   cluster1-04   e3a
true
        cluster1-04_clus2 up/up     169.254.1.7/23   cluster1-04   e3a
false
8 entries were displayed.
cluster1::*>
```

1. [[step9]驗證叢集是否正常：「叢集顯示」

```
cluster1::*> cluster show
Node                  Health  Eligibility   Epsilon
--------------------- ------- ------------  -------
cluster1-01           true    true          false
cluster1-02           true    true          false
cluster1-03           true    true          true
cluster1-04           true    true          false
4 entries were displayed.
cluster1::*>
```

1. [[step10]清除交換器CS2上的組態並執行基本設定。
   a. 清除組態。此步驟需要主控台連線至交換器。

```
cs2# write erase
Warning: This command will erase the startup-configuration.
Do you wish to proceed anyway? (y/n)  [n] y
cs2# reload
This command will reboot the system. (y/n)?  [n] y
cs2#
```

1. 執行交換器的基本設定。
2. [[step11]使用下列傳輸協定之一、將RCF複製到交換器CS2的bootflash：FTP、TFTP、SFTP或scp。如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列NX-OS命令參考指南"。

   此範例顯示使用TFTP將RCF複製到交換器CS2上的bootflash。

```
cs2# copy tftp: bootflash: vrf management
Enter source filename: Nexus_9336C_RCF_v1.6-Cluster-HA-Breakout.txt
Enter hostname for the tftp server: 172.22.201.50
Trying to connect to tftp server......Connection to Server Established.
TFTP get operation was successful
Copy complete, now saving to disk (please wait)...
```

1. [[step12]套用先前下載至bootflash的RCF。如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列NX-OS命令參考指南"。

   此範例顯示在交換器CS2上安裝的RCF檔案「Nexus _9336C_RCF-v1.6-Cluster-HA-Breakout.txt"。

```
cs2# copy Nexus_9336C_RCF_v1.6-Cluster-HA-Breakout.txt running-config
echo-commands
```

1. [[step13]檢查「show banner motd」命令的橫幅輸出。您必須閱讀並遵循這些指示、以確保交換器的組態和操作正確無誤。

```
cs2# show banner motd
**************************************************************************
*
* NetApp Reference Configuration File (RCF)
*
* Switch   : Nexus N9K-C9336C-FX2
* Filename : Nexus_9336C_RCF_v1.6-Cluster-HA-Breakout.txt
* Date     : 10-23-2020
* Version  : v1.6
*
* Port Usage:
* Ports  1- 3: Breakout mode (4x10G) Intra-Cluster Ports, int e1/1/1-4,
* e1/2/1-4, e1/3/1-4
* Ports  4- 6: Breakout mode (4x25G) Intra-Cluster/HA Ports, int e1/4/1-4,
* e1/5/1-4, e1/6/1-4
* Ports  7-34: 40/100GbE Intra-Cluster/HA Ports, int e1/7-34
* Ports 35-36: Intra-Cluster ISL Ports, int e1/35-36
*
* Dynamic breakout commands:
* 10G: interface breakout module 1 port <range> map 10g-4x
* 25G: interface breakout module 1 port <range> map 25g-4x
*
* Undo breakout commands and return interfaces to 40/100G configuration in
* config mode:
* no interface breakout module 1 port <range> map 10g-4x
* no interface breakout module 1 port <range> map 25g-4x
* interface Ethernet <interfaces taken out of breakout mode>
* inherit port-profile 40-100G
* priority-flow-control mode auto
* service-policy input HA
* exit
*
**************************************************************************
*
```

1. [[step14]確認RCF檔案為正確的更新版本：「How running-config」

   當您檢查輸出以確認您擁有正確的RCF時、請確定下列資訊正確無誤：

   - RCF橫幅
   - 節點和連接埠設定
   - 自訂

     輸出會因站台組態而異。請檢查連接埠設定、並參閱版本說明、以瞭解您安裝的RCF的任何特定變更。

2. 驗證RCF版本和交換器設定是否正確之後、請將執行組態檔複製到啟動組態檔。

   如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列NX-OS命令參考指南"。

```
cs2# copy running-config startup-config
[#########################################] 100% Copy complete
```

1. [[step16]重新開機交換器CS2。您可以在交換器重新開機時忽略節點上報告的「叢集連接埠當機」事件。

```
cs2# reload
This command will reboot the system. (y/n)?  [n] y
```

1. [[step17]套用相同的RCF、並再次儲存執行中的組態。

```
cs2# copy Nexus_9336C_RCF_v1.6-Cluster-HA-Breakout.txt running-config
echo-commands
cs2# copy running-config startup-config
[#########################################] 100% Copy complete
```

1. [[step18]驗證叢集上叢集連接埠的健全狀況。
   a. 驗證叢集中所有節點的e0d連接埠是否正常運作：「network port show -role cluster」（網路連接埠show -role叢集）

```
cluster1::*> network port show -role cluster
Node: cluster1-01
                                                                    Ignore
                                                  Speed(Mbps)  Health   Health
Port    IPspace   Broadcast Domain Link MTU  Admin/Oper   Status   Status
------- --------- ---------------- ---- ---- ------------ -------- ------
e3a     Cluster   Cluster          up   9000 auto/100000  healthy  false
e3b     Cluster   Cluster          up   9000 auto/100000  healthy  false

Node: cluster1-02
                                                                    Ignore
                                                   Speed(Mbps)  Health  Health
Port    IPspace   Broadcast Domain Link MTU   Admin/Oper   Status  Status
------- --------- ---------------- ---- ----- ------------ -------- ------
e3a    Cluster   Cluster          up   9000  auto/100000  healthy  false
e3b    Cluster   Cluster          up   9000  auto/100000  healthy  false

Node: cluster1-03
                                                                    Ignore
                                                   Speed(Mbps) Health   Health
Port   IPspace    Broadcast Domain Link MTU   Admin/Oper  Status   Status
------ ---------- ---------------- ---- ----- ----------- -------- ------
e3a    Cluster    Cluster          up   9000  auto/100000 healthy  false
e0d    Cluster    Cluster          up   9000  auto/100000 healthy  false

Node: cluster1-04
                                                                    Ignore
                                                   Speed(Mbps) Health   Health
Port   IPspace    Broadcast Domain Link MTU   Admin/Oper  Status   Status
------ ---------- ---------------- ---- ----- ----------- -------- ------
e3a    Cluster      Cluster        up   9000  auto/100000 healthy  false
e0d    Cluster      Cluster        up   9000  auto/100000 healthy  false
8 entries were displayed.
```

1. 驗證叢集的交換器健全狀況（這可能不會顯示交換器CS2、因為LIF不是位於e0d上）。

```
cluster1::*> network device-discovery show -protocol cdp
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------- --------
cluster1-01/cdp
            e3a    cs1                       Ethernet1/7       N9K-C9336C
            e0d    cs2                       Ethernet1/7       N9K-C9336C
cluster01-2/cdp
            e3a    cs1                       Ethernet1/8       N9K-C9336C
            e0d    cs2                       Ethernet1/8       N9K-C9336C
cluster01-3/cdp
            e3a    cs1                       Ethernet1/1/1     N9K-C9336C
            e3b    cs2                       Ethernet1/1/1     N9K-C9336C
cluster1-04/cdp
            e3a    cs1                       Ethernet1/1/2     N9K-C9336C
            e3b    cs2                       Ethernet1/1/2     N9K-C9336C
cluster1::*> system cluster-switch show -is-monitoring-enabled-operational
true
Switch                      Type               Address          Model
--------------------------- ------------------ ---------------- -----
cs1                         cluster-network    10.233.205.90    NX9-C9336C
     Serial Number: FOCXXXXXXGD
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
    Version Source: CDP
cs2                         cluster-network    10.233.205.91    NX9-C9336C
     Serial Number: FOCXXXXXXGS
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
    Version Source: CDP
2 entries were displayed.
```

您可能會在CS1交換器主控台觀察下列輸出、視先前載入交換器的RCF版本而定。

```
2020 Nov 17 16:07:18 cs1 %$ VDC-1 %$ %STP-2-UNBLOCK_CONSIST_PORT:
Unblocking port port-channel1 on VLAN0092. Port consistency restored.
2020 Nov 17 16:07:23 cs1 %$ VDC-1 %$ %STP-2-BLOCK_PVID_PEER: Blocking
port-channel1 on VLAN0001. Inconsistent peer vlan.
2020 Nov 17 16:07:23 cs1 %$ VDC-1 %$ %STP-2-BLOCK_PVID_LOCAL: Blocking
port-channel1 on VLAN0092. Inconsistent local vlan.
```

1. [[step19]在叢集交換器CS1上、關閉連接至節點叢集連接埠的連接埠。下列範例使用步驟1的介面輸出範例：

```
cs1(config)# interface eth1/1/1-2,eth1/7-8
cs1(config-if-range)# shutdown
```

1. [step20]確認叢集LIF已移轉至交換器CS2上裝載的連接埠。這可能需要幾秒鐘的時間：「網路介面show -role cluster」

```
cluster1::*> network interface show -role cluster
        Logical            Status      Network         Current
Current Is
Vserver Interface          Admin/Oper  Address/Mask    Node         Port
Home
------- ------------------ ----------- --------------- ------------
------- ----
Cluster
        cluster1-01_clus1  up/up       169.254.3.4/23   cluster1-01   e0d
false
        cluster1-01_clus2  up/up       169.254.3.5/23   cluster1-01   e0d
true
        cluster1-02_clus1  up/up       169.254.3.8/23   cluster1-02   e0d
false
        cluster1-02_clus2  up/up       169.254.3.9/23   cluster1-02   e0d
true
        cluster1-03_clus1  up/up       169.254.1.3/23   cluster1-03   e3b
false
        cluster1-03_clus2  up/up       169.254.1.1/23   cluster1-03   e3b
true
        cluster1-04_clus1  up/up       169.254.1.6/23   cluster1-04   e3b
false
        cluster1-04_clus2  up/up       169.254.1.7/23   cluster1-04   e3b
true
8 entries were displayed.
cluster1::*>
```

1. [[step21]驗證叢集是否正常：「叢集顯示」

```
cluster1::*> cluster show
Node                  Health   Eligibility   Epsilon
--------------------- -------- ------------- -------
cluster1-01           true     true          false
cluster1-02           true     true          false
cluster1-03           true     true          true
cluster1-04           true     true          false
4 entries were displayed.
cluster1::*>
```

1. [[step22]重複交換器CS1上的步驟7至14。
2. 在叢集生命體上啟用自動還原。

```
cluster1::*> network interface modify -vserver Cluster -lif * -auto-revert
True
```

1. [[step24]重新開機交換器CS1。您可以這樣做、觸發叢集生命期以恢復到其主連接埠。您可以在交換器重新開機時忽略節點上報告的「叢集連接埠當機」事件。

```
cs1# reload
This command will reboot the system. (y/n)?  [n] y
```

1. [[step25]確認連接至叢集連接埠的交換器連接埠是否正常運作。

```
cs1# show interface brief | grep up
.
.
Eth1/1/1      1       eth  access up      none                   100G(D)
--
Eth1/1/2      1       eth  access up      none                   100G(D)
--
Eth1/7        1       eth  trunk  up      none                   100G(D)
--
Eth1/8        1       eth  trunk  up      none                   100G(D)
--
.
.
```

1. [[step26]確認CS1與CS2之間的ISL正常運作：「How port-channel Summary」（顯示連接埠通道摘要）

```
cs1# show port-channel summary
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------------
------
Group Port-       Type     Protocol  Member Ports      Channel
--------------------------------------------------------------------------
------
1     Po1(SU)     Eth      LACP      Eth1/35(P)        Eth1/36(P)
cs1#
```

1. [[step27]確認叢集lifs已還原至其主連接埠：「網路介面show -role cluster」

```
cluster1::*> network interface show -role cluster
        Logical            Status     Network           Current
Current Is
Vserver Interface          Admin/Oper Address/Mask      Node        Port
Home
------- ------------------ ---------- ----------------- -----------
------- ----
Cluster
        cluster1-01_clus1  up/up      169.254.3.4/23    cluster1-01  e0d
true
        cluster1-01_clus2  up/up      169.254.3.5/23    cluster1-01  e0d
true
        cluster1-02_clus1  up/up      169.254.3.8/23    cluster1-02  e0d
true
        cluster1-02_clus2  up/up      169.254.3.9/23    cluster1-02  e0d
true
        cluster1-03_clus1  up/up      169.254.1.3/23    cluster1-03  e3b
true
        cluster1-03_clus2  up/up      169.254.1.1/23    cluster1-03  e3b
true
        cluster1-04_clus1  up/up      169.254.1.6/23    cluster1-04  e3b
true
        cluster1-04_clus2  up/up      169.254.1.7/23    cluster1-04  e3b
true
8 entries were displayed.
cluster1::*>
```

1. [[step28]驗證叢集是否正常：「叢集顯示」

```
cluster1::*> cluster show
Node                  Health  Eligibility   Epsilon
--------------------- ------- ------------- -------
cluster1-01           true    true          false
cluster1-02           true    true          false
cluster1-03           true    true          true
cluster1-04           true    true          false
4 entries were displayed.
cluster1::*>
```

1. [[step29] Ping遠端叢集介面以驗證連線能力：「叢集ping叢集節點local」

```
cluster1::*> cluster ping-cluster -node local
Host is cluster1-03
Getting addresses from network interface table...
Cluster cluster1-03_clus1 169.254.1.3 cluster1-03 e3a
Cluster cluster1-03_clus2 169.254.1.1 cluster1-03 e3b
Cluster cluster1-04_clus1 169.254.1.6 cluster1-04 e3a
Cluster cluster1-04_clus2 169.254.1.7 cluster1-04 e3b
Cluster cluster1-01_clus1 169.254.3.4 cluster1-01 e3a
Cluster cluster1-01_clus2 169.254.3.5 cluster1-01 e0d
Cluster cluster1-02_clus1 169.254.3.8 cluster1-02 e3a
Cluster cluster1-02_clus2 169.254.3.9 cluster1-02 e0d
Local = 169.254.1.3 169.254.1.1
Remote = 169.254.1.6 169.254.1.7 169.254.3.4 169.254.3.5 169.254.3.8
169.254.3.9
Cluster Vserver Id = 4294967293
Ping status:
............
Basic connectivity succeeds on 12 path(s)
Basic connectivity fails on 0 path(s)
................................................
Detected 9000 byte MTU on 12 path(s):
    Local 169.254.1.3 to Remote 169.254.1.6
    Local 169.254.1.3 to Remote 169.254.1.7
    Local 169.254.1.3 to Remote 169.254.3.4
    Local 169.254.1.3 to Remote 169.254.3.5
    Local 169.254.1.3 to Remote 169.254.3.8
    Local 169.254.1.3 to Remote 169.254.3.9
    Local 169.254.1.1 to Remote 169.254.1.6
    Local 169.254.1.1 to Remote 169.254.1.7
    Local 169.254.1.1 to Remote 169.254.3.4
    Local 169.254.1.1 to Remote 169.254.3.5
    Local 169.254.1.1 to Remote 169.254.3.8
    Local 169.254.1.1 to Remote 169.254.3.9
Larger than PMTU communication succeeds on 12 path(s)
RPC status:
6 paths up, 0 paths down (tcp check)
6 paths up, 0 paths down (udp check)
```

在Cisco Nexus 9336C-FX2儲存交換器上安裝RCF

參考組態檔（RCF）可在Cisco Nexus 9336C-FX2儲存交換器上升級。

在升級儲存交換器上的RCF之前、必須符合下列條件：

- 交換器必須完全正常運作（記錄中不應有錯誤或類似問題）。

- 如果您只安裝NX-OS並保留目前的RCF版本、則必須在RCF中檢查或設定所需的開機變數、以反映所需的開機映像。
- 如果您需要變更開機變數以反映目前的開機映像、則必須在重新套用RCF之前執行此動作、以便在未來重新開機時產生正確的版本。
- 您必須參閱Cisco網站上提供的適當軟體與升級指南、以取得Cisco儲存設備升級與降級程序的完整文件。請參閱 "Cisco Nexus 9000系列交換器" 以取得更多資訊。
- 100個GbE連接埠的數量是在上可用的參考組態檔（RCT）中定義 "Cisco乙太網路交換器" 頁面。

程序摘要

1. 檢查交換器和連接埠的健全狀況（步驟1-4）
2. 將NX-OS映像下載至Cisco交換器ST2並重新開機（步驟5至8）
3. 將RCF複製到Cisco交換器ST2（步驟9-12）
4. 重新檢查交換器和連接埠的健全狀況（步驟13-15）
5. 對Cisco交換器ST1重複步驟1-15。

> 命令輸出可能會因ONTAP 不同版本的不相同而有所差異。

本程序中的範例使用下列交換器和節點命名法：

- 兩個儲存交換器的名稱為_ST1_和_ST2_。
- 節點是_node1_和_node2_。

> 此程序需要同時使用ONTAP 支援指令和Cisco Nexus 9000系列交換器的指令；ONTAP 除非另有說明、否則會使用支援指令。

步驟

1. 如果AutoSupport 此叢集啟用了「支援功能」、請叫用AutoSupport 下列訊息來抑制自動建立案例：「System Node AutoSupport 支援功能叫用節點*-type all - Message MAn=xh」

   其中x是維護時段的持續時間（以小時為單位）。

2. 檢查儲存交換器是否可用：「System switches Ethernet show」

```
storage::*> system switch ethernet show
Switch                    Type               Address          Model
------------------------- ------------------ ----------------
---------------
st1
                          storage-network    172.17.227.5     NX9-C9336C
      Serial Number: FOC221206C2
       Is Monitored: true
             Reason: None
   Software Version: Cisco Nexus Operating System (NX-OS) Software,
Version
                     9.3(5)
     Version Source: CDP
st2
                          storage-network    172.17.227.6     NX9-C9336C
      Serial Number: FOC220443LZ
       Is Monitored: true
             Reason: None
   Software Version: Cisco Nexus Operating System (NX-OS) Software,
Version
                     9.3(5)
     Version Source: CDP
2 entries were displayed.
storage::*>
```

1. [[step3]驗證節點連接埠是否正常運作：「torage port show -port-type ENET」

```
storage::*> storage port show -port-type ENET
                                   Speed                             VLAN
Node      Port   Type    Mode       (Gb/s)     State     Status        ID
------- ------- ------- ---------- ---------- --------- ----------- -----
node1
        e3a     ENET    storage    100        enabled   online         30
        e3b     ENET    storage      0        enabled   offline        30
        e7a     ENET    storage      0        enabled   offline        30
        e7b     ENET    storage    100        enabled   online         30
node2
        e3a     ENET    storage    100        enabled   online         30
        e3b     ENET    storage      0        enabled   offline        30
        e7a     ENET    storage      0        enabled   offline        30
        e7b     ENET    storage    100        enabled   online         30
```

1. 檢查叢集是否有儲存交換器或纜線問題：「System health alerting show -instance」（系統健全狀況警示顯示-instance）

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step5]下載NX-OS映像以切換ST2。
2. 安裝系統映像、以便下次重新啟動交換器ST2時載入新版本。交換器會在10秒內重新開機、並顯示新映像、如下面的輸出所示：

```
st2# install all nxos bootflash:nxos.9.3. 5.bin
Installer will perform compatibility check first. Please wait.
Installer is forced disruptive
Verifying image bootflash:/nxos.9.3.4.bin for boot variable "nxos".
[####################] 100% -- SUCCESS
Verifying image type.
[[####################] 100% -- SUCCESS
Preparing "nxos" version info using image bootflash:/nxos.9.3.4.bin.
[####################] 100% -- SUCCESS
Preparing "bios" version info using image bootflash:/nxos.9.3.4.bin.
[####################] 100% -- SUCCESS
Performing module support checks.
[####################] 100% -- SUCCESS
Notifying services about system upgrade.
[####################] 100% -- SUCCESS
Compatibility check is done:
Module  bootable  Impact  Install-type  Reason
------    --------   ----- --------   ------------   ---- --
     1        yes      disruptive         reset  default upgrade is not
hitless
Images will be upgraded according to following table:
Module Image        Running-Version(pri:alt)               New-Version
Upg

Required
------ --------  ----------------------------------------  ------------
--------
 1     nxos                                       9.3(3)          9.3(4)
yes
 1     bios      v08.37(01/28/2020):v08.23(09/23/2015)
v08.38(05/29/2020)   no
Switch will be reloaded for disruptive upgrade.
Do you want to continue with the installation (y/n)? [n] y
input string too long
Do you want to continue with the installation (y/n)? [n] y
Install is in progress, please wait.
Performing runtime checks.
```

```
[####################] 100% -- SUCCESS
Setting boot variables.
[####################] 100% -- SUCCESS
Performing configuration copy.
[####################] 100% -- SUCCESS
Module 1: Refreshing compact flash and upgrading bios/loader/bootrom.
Warning: please do not remove or power off the module at this time.
[####################] 100% -- SUCCESS
Finishing the upgrade, switch will reboot in 10 seconds.
st2#
```

1. [[step7]儲存組態。

   系統會提示您重新開機、如下例所示：

```
st2# copy running-config startup-config
[########################################] 100% Copy complete.
st2# reload
This command will reboot the system. (y/n)? [n] y
```

1. [[step8]確認交換器上有新的NX-OS版本號碼。

```
st2# show version
Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
Upgrading a Cisco Nexus 9336C Storage Switch 6
Upgrading a Cisco Nexus 9336C storage switch
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A copy of each such license is available at
http://www.opensource.org/licenses/gpl-2.0.php and
http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
.
Software
 BIOS: version 08.38
 NXOS: version 9.3(5)
 BIOS compile time: 05/29/2020
 NXOS image file is: bootflash:///nxos.9.3. 5.bin
 NXOS compile time: 4/28/2020 21:00:00 [04/29/2020 02:28:31]
Hardware
 cisco Nexus9000 C9336C Chassis (Nexus 9000 Series)
 Intel(R) Xeon(R) CPU E5-2403 v2 @ 1.80GHz with 8154432 kB of memory.
 Processor Board ID FOC20291J6K
 Device name: S2
 bootflash: 53298520 kB
Kernel uptime is 0 day(s), 0 hour(s), 3 minute(s), 42 second(s)
Last reset at 157524 usecs after Mon Nov 2 18:32:06 2020
           Reason: Reset due to upgrade
   System version: 9.3(5)
   Service:
plugin
   Core Plugin, Ethernet Plugin
   Active Package(s):
st2#
```

1. 使用下列傳輸協定之一、將交換器ST2上的RCF複製到交換器bootflash：FTP、HTTP、TFTP、SFTP或scp。

   如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列NX-OS命令參考指南"。

   以下範例顯示HTTP用於將RCF複製到交換器ST2上的bootflash：

```
st2# copy http://172.16.10.1//cfg/Nexus_9336C_RCF_v1.6-Storage.txt
bootflash: vrf management
% Total % Received % Xferd   Average   Speed  Time   Time   Time
Current
   Dload     Upload  Total Spent   Left
Speed
 100    3254          100     3254      0       0     8175    0 --:--:--
--:--:-- --:--:-
8301
Copy complete, now saving to disk (please wait)...
Copy complete.
st2#
```

1. [[step10]套用先前下載至bootflash的RCF：「copy bootflash」。

   以下範例顯示在交換器ST2上安裝的RCF檔案「Nexus _9336C_RCF-v1.6-Storage.txt」：

```
st2# copy Nexus_9336C_RCF_v1.6-Storage.txt running-config echo-commands
```

1. [[step11]確認RCF檔案為正確的更新版本：「How running-config」

   當您檢查輸出以確認您擁有正確的RCF時、請確定下列資訊正確無誤：

   - RCF橫幅
   - 節點和連接埠設定
   - 自訂

     輸出會因站台組態而異。請檢查連接埠設定、並參閱版本說明、以瞭解您安裝的RCF的任何特定變更。

   *重要事項：*在「show banner motd」命令的橫幅輸出中、您必須閱讀並遵循*重要事項*一節中的指示、以確保交換器的組態和操作正確無誤。

```
st2# show banner motd
**************************************************************************
****
*NetApp Reference Configuration File (RCF)
*
*Switch : Nexus N9K-C9336C-FX2
*Filename : Nexus_9336C_RCF_v1.6-Storage.txt
* Date : 10-23-2020
*Version  : v1.6
*
*Port Usage: Storage configuration
*Ports 1-36: 100GbE Controller and Shelf Storage Ports
*
*IMPORTANT NOTES*
*- This RCF utilizes QoS and requires TCAM re-configuration, requiring RCF
*to be loaded twice with the Storage Switch rebooted in between.
*
*- Perform the following 4 steps to ensure proper RCF installation:
*
*(1) Apply RCF first time, expect following messages:
*- Please save config and reload the system...
*- Edge port type (portfast) should only be enabled on ports...
*- TCAM region is not configured for feature QoS class IPv4 ingress...
*
*(2) Save running-configuration and reboot Cluster Switch
*
*(3) After reboot, apply same RCF second time and expect following
messages:
*- % Invalid command at '^' marker
*- Syntax error while parsing...
*
*(4) Save running-configuration again
**************************************************************************
****
st2#
```

1. [[step12]驗證軟體版本和交換器設定是否正確之後、請將執行組態檔複製到交換器ST2上的啟動組態檔。

   如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列NX-OS命令參考指南"。

   以下範例顯示「執行組態」檔案已成功複製到「儲存組態」檔案：

```
st2# copy running-config startup-config
[#######################################] 100% Copy complete.
```

1. [[step13]重新檢查儲存交換器在重新開機後是否可用：「System switches Ethernet show」

```
storage::*> system switch ethernet show
Switch                       Type               Address          Model
---------------------------- ------------------ ----------------
---------------
st1
                             storage-network     172.17.227.5     NX9-
C9336C
     Serial Number: FOC221206C2
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
    Version Source: CDP
st2
                             storage-network      172.17.227.6    NX9-
C9336C
     Serial Number: FOC220443LZ
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
    Version Source: CDP
2 entries were displayed.
storage::*
```

1. [[step14]重新開機後、請確認交換器連接埠正常運作：「torage port show -port-type ENET」

```
storage::*> storage port show -port-type ENET
                                  Speed                             VLAN
Node    Port    Type    Mode       (Gb/s)       State     Status       ID
------- ------- ------- ---------- ----------- --------- --------- -----
node1
        e3a     ENET    storage           100   enabled   online       30
        e3b     ENET    storage             0   enabled   offline      30
        e7a     ENET    storage             0   enabled   offline      30
        e7b     ENET    storage           100   enabled   online       30
node2
        e3a     ENET    storage           100   enabled   online       30
        e3b     ENET    storage             0   enabled   offline      30
        e7a     ENET    storage             0   enabled   offline      30
        e7b     ENET    storage           100   enabled   online       30
```

1. 重新檢查叢集沒有儲存交換器或纜線問題：「系統健全狀況警示顯示-instance」

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step16]重複此程序、即可在交換器ST1上使用RCF。
2. 如果您禁止自動建立個案、請叫用AutoSupport 下列訊息重新啟用此功能：「System Node AutoSupport 現象叫用節點*-type all -most MAn=end」

在Cisco Nexus 9336C-FX2共享交換器上安裝RCF

從S299.1開始ONTAP 、您可以使用Cisco Nexus 9336C-FX2交換器、將儲存設備和叢集功能結合到共享交換器情境中。

開始之前

- 叢集交換器必須完全正常運作（記錄中不應有錯誤或類似問題）。
- 儲存交換器必須完全正常運作（記錄中不應有錯誤或類似問題）。
- 這兩個儲存交換器的名稱為_SH1_和_sh2_。
- 此處使用的範例將共用RCF載入新交換器。

步驟

1. 使用下列傳輸傳輸協定之一、將交換器sh2上的RCF複製到交換器bootflash：FTP、HTTP、TFTP、SFTP或scp。

   如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列NX-OS命令參考指南"。

   下列範例顯示HTTP正用於將RCF複製到交換器sh2上的bootflash：

```
sh2# copy http://172.16.10.1//cfg/Nexus_9336C_RCF_v1.7-Cluster-Ha-
Storage.txt bootflash: vrf management
% Total % Received % Xferd   Average   Speed  Time   Time   Time
Current
   Dload     Upload  Total Spent   Left
Speed
 100    5143          100     5143      0       0     11300    0 --:--:--
--:--:-- --:--:-
11300
Copy complete, now saving to disk (please wait)...
Copy complete.
sh2#
```

1. 套用先前下載至bootflash的RCF：「copy bootflash」。

   以下範例顯示在交換器sh2上安裝的RCF檔案「Nexus_9336C_RCF_v1.7-Cluster-HA-Storage.txt」：

```
sh2# copy Nexus_9336C_RCF_v1.7-Cluster-HA-Storage.txt running-config echo-
commands
```

1. 確認RCF檔案為正確的更新版本：「How running-config」

   當您檢查輸出以確認您擁有正確的RCF時、請確定下列資訊正確無誤：

   - RCF橫幅
   - 節點和連接埠設定
   - 自訂

     輸出會因站台組態而異。請檢查連接埠設定、並參閱版本說明、以瞭解您安裝的RCF的任何特定變更。

   *重要事項：*在「show banner motd」命令的橫幅輸出中、您必須閱讀並遵循*重要事項*一節中的指示、以確保交換器的組態和操作正確無誤。

```
sh2# show banner motd
**************************************************************************
****
*NetApp Reference Configuration File (RCF)
*
*Switch : Nexus N9K-C9336C-FX2
*Filename : Nexus_9336C_RCF_v1.7-Cluster-HA-Storage.txt
* Date :  Jan-08-2021
*Version  : v1.7
*
*Port Usage:
*Ports  1-8: 40/100GbE Intra-Cluster/HA Ports, int e1/1-8
*Port     9: 10GbE breakout Intra-Cluster Ports, int e1/9/1-4
*Port    10: 25GbE breakout Intra-Cluster/HA Ports, int e1/10/1-4
*Ports 11-22: First HA-pair Controller and Shelf Storage Ports, int e1/11-
22
*Ports 23-34: Second HA-pair Controller and Shelf Storage Ports, int
e1/23-34
*Ports 35-36: Intra-Cluster ISL Ports, int e1/35-36
*
* Undo breakout commands and return interfaces to 40/100G configuration in
* config mode:
* no interface breakout module 1 port 9 map 10g-4x
* no interface breakout module 1 port 10 map 25g-4x
* interface Ethernet 1/9-10
* inherit port-profile CLUSTER_HA
* priority-flow-control mode auto
* service-policy type qos input HA_POLICY
* exit
*
*IMPORTANT NOTES*
* In certain conditions, N9K-C9336C-FX2 may not be able to auto-negotiate
port
* speed correctly, and port speed must be manually set, in config mode,
e.g.
* int e1/1
* speed 40000
* int e1/3
* speed 100000
*
**************************************************************************
****
sh2#
```

1. 驗證軟體版本和交換器設定是否正確之後、請將「執行組態」檔案複製到交換器sh2上的啟動組態檔。

如需Cisco命令的詳細資訊、請參閱中的適當指南 "Cisco Nexus 9000系列NX-OS命令參考指南"。

以下範例顯示「執行組態」檔案已成功複製到「儲存組態」檔案：

```
sh2# copy running-config startup-config
[#######################################] 100% Copy complete.
```

1. 重複此程序、即可在交換器SH1上使用RCF。

透過新增兩個共用交換器、從具有直接附加儲存設備的無交換器叢集移轉

使用直接附加儲存設備從無交換器叢集移轉

將雙節點無交換式叢集（不中斷營運）移轉至採用Cisco Nexus 9336C-FX2叢集交換器的叢集時、您必須瞭解某些組態資訊、連接埠連線和纜線需求。您使用的程序取決於每個控制器上是否有兩個專屬的叢集網路連接埠、或每個控制器上是否有一個叢集連接埠。記錄的程序適用於所有使用光纖或雙軸纜線連接埠的節點、但如果節點使用內建10Gb Base-T RJ45連接埠作為叢集網路連接埠、則此交換器不支援此程序。

大多數系統在每個控制器上都需要兩個專用的叢集網路連接埠。請參閱 "Cisco乙太網路交換器" 以取得更多資訊。

如果您現有的雙節點無交換式叢集環境、可以使用Cisco Nexus 9336C-FX2交換器移轉至雙節點交換式叢集環境、以便擴充至叢集中的兩個節點以外。

開始之前

- 雙節點無交換器組態：
  - 雙節點無交換器組態必須正確設定並正常運作。
  - 節點必須執行ONTAP 的是更新版本的版本。
  - 所有叢集連接埠都必須處於* up *狀態。
  - 所有叢集邏輯介面（lifs）都必須處於* up 狀態、而且必須位於 home*連接埠上。
- Cisco Nexus 9336C-FX2交換器組態：
  - 兩台交換器都必須具備管理網路連線能力。
  - 叢集交換器必須有主控台存取權。
  - Nexus 9336C-FX2節點對節點交換器和交換器對交換器連線必須使用雙軸纜線或光纖纜線。
  - NetApp "Hardware Universe" 包含纜線的詳細資訊。
  - 交換器間連結（ISL）纜線必須連接至9336C-FX2交換器的連接埠1/35和1/36。
- 9336C-FX2交換器的初始自訂必須完成。因此：
  - 9336C-FX2交換器正在執行最新版本的軟體
  - 參考組態檔（RCT）已套用至交換器
  - 必須在新交換器上設定任何站台自訂、例如：SMTP、SNMP和SSH。

本程序中的範例使用下列叢集交換器和節點命名法：

- 9336C-FX2交換器的名稱為_CS1_和_CS2_。
- 叢集SVM的名稱為_node1_和_node2_。
- lifs的名稱分別是節點1上的_node1_clus1_和節點1_clus2_、節點2上的節點2_clus1_和節點2_clus2_。
- cluster1：：*>提示會指出叢集的名稱。
- 此程序中使用的叢集連接埠是_e3a_和_e3c_、如同AFF 根據ESIA400控制器。。 "Hardware Universe" 包含平台實際叢集連接埠的最新資訊。

步驟

1. 如果AutoSupport 此叢集啟用了「支援功能」、請叫用AutoSupport 「系統節點AutoSupport 支援功能叫用節點*-type all -most MAn=xh」訊息來禁止自動建立案例。

   其中x是維護時段的持續時間（以小時為單位）。

   ⓘ 此資訊可通知技術支援人員執行此維護工作、以便在維護期間抑制自動建立案例。AutoSupport

1. 將權限層級變更為進階、並在系統提示您繼續時輸入y：「et -priv榮幸advanced'（設定-權限進階）」

   出現進階提示（*>）。

2. 在新的叢集交換器CS1和CS2上停用所有面向節點的連接埠（非ISL連接埠）。您不得停用ISL連接埠。

   以下範例顯示、交換器CS1上的節點對介連接埠1至34已停用：

```
cs1# config
Enter configuration commands, one per line. End with CNTL/Z.
cs1(config)# interface e/1-34
cs1(config-if-range)# shutdown
```

1. [[step4]驗證兩個9336C-FX2交換器CS1和CS2之間的ISL和實體連接埠是否在連接埠1/35和1/36上為開啟狀態：「顯示連接埠通道摘要」

   下列範例顯示交換器CS1上的ISL連接埠已開啟：

```
cs1# show port-channel summary
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------------
------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------------
------
1     Po1(SU)     Eth      LACP      Eth1/35(P)   Eth1/36(P)
```

下列範例顯示交換器CS2上的ISL連接埠已開啟：

```
       cs2# show port-channel summary
        Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------------
------
Group Port-       Type     Protocol  Member Ports
      Channel
--------------------------------------------------------------------------
------
1     Po1(SU)     Eth      LACP      Eth1/35(P)   Eth1/36(P)
```

1. [[step5]顯示鄰近裝置的清單：「How cup neighbor」。

   此命令提供有關連線至系統之裝置的資訊。下列範例列出交換器CS1上的鄰近裝置：

```
cs1# show cdp neighbors
Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute
Device-ID          Local Intrfce  Hldtme Capability  Platform      Port ID
cs2                Eth1/35        175    R S I s     N9K-C9336C    Eth1/35
cs2                Eth1/36        175    R S I s     N9K-C9336C    Eth1/36
Total entries displayed: 2
```

下列範例列出交換器CS2上的鄰近裝置：

```
cs2# show cdp neighbors
Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute
Device-ID          Local Intrfce  Hldtme Capability  Platform      Port ID
cs1                Eth1/35        177    R S I s     N9K-C9336C    Eth1/35
cs1           )    Eth1/36        177    R S I s     N9K-C9336C    Eth1/36

Total entries displayed: 2
```

1. [[step6]確認所有叢集連接埠都已啟動：「network port show - IPSpace Cluster（網路連接埠顯示- IPSpace 叢集）」

   每個連接埠應顯示「Link」（連結）和「Healthy for Health」（健全狀況）狀態：

```
cluster1::*> network port show -ipspace Cluster

Node: node1
                                                        Speed(Mbps)  Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
--------- ------------ ---------------- ---- ---- ------------ ---------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
e3b       Cluster      Cluster          up   9000  auto/100000 healthy

Node: node2
                                                        Speed(Mbps)  Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
--------- ------------ ---------------- ---- ---- ------------ ---------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
e3b       Cluster      Cluster          up   9000  auto/100000 healthy
4 entries were displayed.
```

1. [[step7]確認所有叢集生命週年都在運作中：「network interface show - vserver cluster」（網路介面顯示-vserver叢集）

   每個叢集LIF都應該顯示為「原位」、並具有「狀態管理/作業者」的開機/開機狀態。

```
cluster1::*> network interface show -vserver Cluster
            Logical     Status     Network            Current
Current Is
Vserver     Interface   Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ ------------- -------
-----
Cluster
            node1_clus1  up/up    169.254.209.69/16  node1         e3a
true
            node1_clus2  up/up    169.254.49.125/16  node1         e3b
true
            node2_clus1  up/up    169.254.47.194/16  node2         e3a
true
            node2_clus2  up/up    169.254.19.183/16  node2         e3b
true
4 entries were displayed.
```

1. [[step8]確認所有叢集生命體上都已啟用自動還原功能：「network interface show - vserver cluster -Fields autosupgres'（網路介面顯示- Vserver叢集-欄位自動還原）」

```
cluster1::*> network interface show -vserver Cluster -fields auto-revert
       Logical
Vserver   Interface     Auto-revert
--------- ------------- ------------
Cluster
          node1_clus1   true
          node1_clus2   true
          node2_clus1   true
          node2_clus2   true
4 entries were displayed.
```

1. [[step9]使用936C-FX2交換器所支援的適當纜線、從節點1上的叢集連接埠e3a拔下纜線、然後將e3a連接至叢集交換器CS1上的連接埠1。

   NetApp "Hardware Universe" 包含纜線的詳細資訊。

2. 從節點2上的叢集連接埠e3a拔下纜線、然後使用9336C-FX2交換器支援的適當纜線、將e3a連接至叢集交換器CS1上的連接埠2。
3. 在叢集交換器CS1上啟用所有面向節點的連接埠。

   下列範例顯示交換器CS1上已啟用連接埠1/1至1/34：

```
cs1# config
Enter configuration commands, one per line. End with CNTL/Z.
cs1(config)# interface e1/1-34
cs1(config-if-range)# no shutdown
```

1. [[step12]驗證所有叢集生命體是否均為*正常運作*、並顯示為真實的「is Home」：「network interface show - vserver Cluster（網路介面顯示- Vserver叢集）」

   以下範例顯示節點1和節點2上的所有lifs均為* up 、且「**is Home**」結果為 true*：

```
cluster1::*> network interface show -vserver Cluster
          Logical      Status     Network            Current     Current
Is
Vserver   Interface    Admin/Oper Address/Mask       Node        Port
Home
--------- ------------ ---------- ------------------ ----------- -------
----
Cluster
          node1_clus1  up/up      169.254.209.69/16  node1       e3a
true
          node1_clus2  up/up      169.254.49.125/16  node1       e3b
true
          node2_clus1  up/up      169.254.47.194/16  node2       e3a
true
          node2_clus2  up/up      169.254.19.183/16  node2       e3b
true
4 entries were displayed.
```

1. [[step13]顯示叢集中節點狀態的相關資訊：「叢集顯示」

   下列範例顯示叢集中節點的健全狀況和資格資訊：

```
cluster1::*> cluster show
Node                 Health  Eligibility   Epsilon
-------------------- ------- ------------  ------------
node1                true    true          false
node2                true    true          false
2 entries were displayed.
```

1. [[step14]使用9336C-FX2交換器支援的適當纜線、從節點1上的叢集連接埠e3b拔下纜線、然後將e3b連接至叢集交換器CS2上的連接埠1。
2. 從節點2上的叢集連接埠e3b拔下纜線、然後使用9336C-FX2交換器支援的適當纜線、將e3b連接至叢集交換器CS2上的連接埠2。
3. 在叢集交換器CS2上啟用所有面向節點的連接埠。

   下列範例顯示交換器CS2上已啟用連接埠1/1至1/34：

```
cs2# config
Enter configuration commands, one per line. End with CNTL/Z.
cs2(config)# interface e1/1-34
cs2(config-if-range)# no shutdown
```

1. [[step17]確認所有叢集連接埠都已啟動：「network port show - IPSpace Cluster（網路連接埠顯示- IPSpace叢集）」

   以下範例顯示節點1和節點2上的所有叢集連接埠都已啟動：

```
cluster1::*> network port show -ipspace Cluster

Node: node1

Ignore
                                                     Speed(Mbps)  Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
Status
--------- ------------ ---------------- ---- ---- ------------ --------
------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
false
e3b       Cluster      Cluster          up   9000  auto/100000 healthy
false

Node: node2

Ignore
                                                     Speed(Mbps)  Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
Status
--------- ------------ ---------------- ---- ---- ------------ --------
------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
false
e3b       Cluster      Cluster          up   9000  auto/100000 healthy
false
4 entries were displayed.
```

1. [[step18]驗證所有介面是否都顯示為「is Home」（原為首頁）：「network interface show - vserver Cluster（網路介面顯示- Vserver叢集）」

   ⓘ 這可能需要幾分鐘的時間才能完成。

以下範例顯示節點1和節點2上的所有lifs均為* up *、且「is Home」結果為真：

```
cluster1::*> network interface show -vserver Cluster
          Logical      Status     Network            Current    Current Is
Vserver   Interface    Admin/Oper Address/Mask       Node       Port
Home
--------- ------------ ---------- ------------------ ---------- -------
----
Cluster
          node1_clus1  up/up      169.254.209.69/16  node1      e3a
true
          node1_clus2  up/up      169.254.49.125/16  node1      e3b
true
          node2_clus1  up/up      169.254.47.194/16  node2      e3a
true
          node2_clus2  up/up      169.254.19.183/16  node2      e3b
true
4 entries were displayed.
```

1. [[step19]確認兩個節點都有一個連線可連線至每個交換器：「How cup neighbor」

   以下範例顯示兩個交換器的適當結果：

```
cs1# show cdp neighbors
Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute
Device-ID          Local Intrfce  Hldtme Capability  Platform      Port ID
node1              Eth1/1         133    H           AFFA400       e3a
node2              Eth1/2         133    H           AFFA400       e3a
cs2                Eth1/35        175    R S I s     N9K-C9336C    Eth1/35
cs2                Eth1/36        175    R S I s     N9K-C9336C    Eth1/36
Total entries displayed: 4
cs2# show cdp neighbors
Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute
Device-ID          Local Intrfce  Hldtme Capability  Platform      Port ID
node1              Eth1/1         133    H           AFFA400       e3b
node2              Eth1/2         133    H           AFFA400       e3b
cs1                Eth1/35        175    R S I s     N9K-C9336C    Eth1/35
cs1                Eth1/36        175    R S I s     N9K-C9336C    Eth1/36
Total entries displayed: 4
```

1. [[step20]顯示叢集中探索到的網路裝置相關資訊：「network device-dDiscovery show -protocol cup」

```
cluster1::*> network device-discovery show -protocol cdp
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------
----------------
node2       /cdp
            e3a    cs1                       0/2               N9K-C9336C
            e3b    cs2                       0/2               N9K-C9336C

node1       /cdp
            e3a    cs1                       0/1               N9K-C9336C
            e3b    cs2                       0/1               N9K-C9336C
4 entries were displayed.
```

1. [[step2a]確認HA配對1（和HA配對2）的儲存組態正確無誤、而且沒有錯誤：「System Switch Ethernet show」（系統交換器乙太網路顯示）

```
storage::*> system switch ethernet show
Switch                    Type                    Address         Model
------------------------- ----------------------- ---------------
----------
sh1
                          storage-network         172.17.227.5    C9336C

       Serial Number: FOC221206C2
        Is Monitored: true
              Reason: None
    Software Version: Cisco Nexus Operating System (NX-OS) Software,
Version
                      9.3(5)
      Version Source: CDP
sh2
                          storage-network         172.17.227.6    C9336C
       Serial Number: FOC220443LZ
        Is Monitored: true
              Reason: None
    Software Version: Cisco Nexus Operating System (NX-OS) Software,
Version
                      9.3(5)
      Version Source: CDP
2 entries were displayed.
storage::*>
```

1. [[step22]確認設定已停用：「network options switchless cluster show」

> 命令可能需要幾分鐘的時間才能完成。等待「3分鐘壽命到期」公告。

以下範例中的「假」輸出顯示組態設定已停用：

```
cluster1::*> network options switchless-cluster show
Enable Switchless Cluster: false
```

1. [[step23]驗證叢集中節點成員的狀態：「cluster show」（叢集顯示）

   下列範例顯示叢集中節點的健全狀況和資格資訊：

```
cluster1::*> cluster show
Node                  Health  Eligibility   Epsilon
--------------------- ------- ------------  --------
node1                 true    true          false
node2                 true    true          false
```

1. [[step24]確保叢集網路具備完整連線能力：「叢集ping叢集節點節點名稱」

```
cluster1::*> cluster ping-cluster -node node2
Host is node2
Getting addresses from network interface table...
Cluster node1_clus1 169.254.209.69 node1 e3a
Cluster node1_clus2 169.254.49.125 node1 e3b
Cluster node2_clus1 169.254.47.194 node2 e3a
Cluster node2_clus2 169.254.19.183 node2 e3b
Local = 169.254.47.194 169.254.19.183
Remote = 169.254.209.69 169.254.49.125
Cluster Vserver Id = 4294967293
Ping status:
....
Basic connectivity succeeds on 4 path(s)
Basic connectivity fails on 0 path(s)
................
Detected 9000 byte MTU on 4 path(s):
Local 169.254.47.194 to Remote 169.254.209.69
Local 169.254.47.194 to Remote 169.254.49.125
Local 169.254.19.183 to Remote 169.254.209.69
Local 169.254.19.183 to Remote 169.254.49.125
Larger than PMTU communication succeeds on 4 path(s)
RPC status:
2 paths up, 0 paths down (tcp check)
2 paths up, 0 paths down (udp check)
```

1. [[step25]將權限層級改回admin：「et -priv. admin」
2. 使用下列命令啟用乙太網路交換器健全狀況監視器記錄收集功能、以收集交換器相關的記錄檔：
    - 「系統交換器乙太網路記錄設定密碼」
    - 「系統交換器乙太網路記錄啟用收集」

```
cluster1::*> system switch ethernet log setup-password
Enter the switch name: <return>
The switch name entered is not recognized.

Choose from the following list:
cs1
cs2
cluster1::*> system switch ethernet log setup-password
Enter the switch name: cs1
RSA key fingerprint is e5:8b:c6:dc:e2:18:18:09:36:63:d9:63:dd:03:d9:cc
Do you want to continue? {y|n}::[n] y
Enter the password: <enter switch password>
Enter the password again: <enter switch password>
cluster1::*> system switch ethernet log setup-password
Enter the switch name: cs2
RSA key fingerprint is 57:49:86:a1:b9:80:6a:61:9a:86:8e:3c:e3:b7:1f:b1
Do you want to continue? {y|n}:: [n] y
Enter the password: <enter switch password>
Enter the password again: <enter switch password>
cluster1::*> system  switch ethernet log enable-collection
Do you want to enable cluster log collection for all nodes in the cluster?
{y|n}: [n] y
Enabling cluster switch log collection.
cluster1::*>
```

設定共享交換器

本程序中的範例使用下列交換器和節點命名法：

- 這兩個共享交換器的名稱為_SH1_和_sh2_。
- 節點是_node1_和_node2_。

> 此程序需要同時使用ONTAP 支援指令和Cisco Nexus 9000系列交換器的指令ONTAP 、除非另有說明、否則會使用指令。

步驟

1. 確認HA配對1（和HA配對2）的儲存組態正確無誤、而且沒有錯誤：「System Switch Ethernet show」（系統交換器乙太網路顯示）

```
storage::*> system switch ethernet show
Switch                      Type                    Address          Model
-------------------------- ---------------------  --------------- -------
sh1
                            storage-network        172.17.227.5    C9336C

      Serial Number: FOC221206C2
       Is Monitored: true
             Reason: None
   Software Version: Cisco Nexus Operating System (NX-OS) Software,
Version
                     9.3(5)
     Version Source: CDP
sh2
                            storage-network        172.17.227.6    C9336C
       Serial Number: FOC220443LZ
        Is Monitored: true
              Reason: None
    Software Version: Cisco Nexus Operating System (NX-OS) Software,
Version
                      9.3(5)
      Version Source: CDP
2 entries were displayed.
storage::*>
```

1. [[step2]確認儲存節點連接埠正常運作：「torage port show -port-type ENET」

```
storage::*> storage port show -port-type ENET
                                    Speed                              VLAN
Node    Port    Type    Mode        (Gb/s)       State      Status       ID
------- ------- ------- ---------- ----------- ---------- ---------- -----
node1
        e0c     ENET   storage          100      enabled  online        30
        e0d     ENET   storage          100      enabled  online        30
        e5a     ENET   storage          100      enabled  online        30
        e5b     ENET   storage          100      enabled  online        30

node2
        e0c     ENET  storage           100      enabled  online        30
        e0d     ENET  storage           100      enabled  online        30
        e5a     ENET  storage           100      enabled  online        30
        e5b     ENET  storage           100      enabled  online        30
```

1. 將HA配對1、NSM224路徑A連接埠移至SH1連接埠範圍11-22。

2. 安裝HA配對1、節點1、SH1連接埠範圍11-22路徑A的纜線。例如AFF 、在不經意的情況下、儲存連接埠的路徑是e0c。
3. 安裝從HA配對1、節點2、路徑A到SH1連接埠範圍11-22的纜線。
4. 驗證節點連接埠是否正常運作：「torage port show -port-type ENET」

```
storage::*> storage port show -port-type ENET
                                    Speed                              VLAN
Node     Port     Type     Mode       (Gb/s)       State      Status      ID
------- ------- ------- ---------- ----------- ---------- ---------- -----
node1
        e0c      ENET   storage          100      enabled  online        30
        e0d      ENET   storage            0      enabled  offline       30
        e5a      ENET   storage            0      enabled  offline       30
        e5b      ENET   storage          100      enabled  online        30

node2
        e0c      ENET  storage           100      enabled  online        30
        e0d      ENET  storage             0      enabled  offline       30
        e5a      ENET  storage             0      enabled  offline       30
        e5b      ENET  storage           100      enabled  online        30
```

1. [[step7]檢查叢集是否有儲存交換器或纜線問題：「系統健全狀況警示顯示-instance」

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step8]將HA配對1、NSM224路徑B連接埠移至sh2連接埠範圍11-22。
2. 安裝從HA配對1、節點1、路徑B到sh2連接埠範圍11-22的纜線。例如AFF 、位於ESIVA400上的路徑B儲存連接埠是e5b。
3. 安裝從HA配對1、節點2、路徑B到sh2連接埠範圍11-22的纜線。
4. 驗證節點連接埠是否正常運作：「torage port show -port-type ENET」

```
storage::*> storage port show -port-type ENET
                                     Speed                              VLAN
Node    Port    Type    Mode        (Gb/s)       State      Status       ID
------- ------- ------- ---------- ----------- ---------- ---------- -----
node1
        e0c     ENET   storage          100      enabled  online        30
        e0d     ENET   storage            0      enabled  offline       30
        e5a     ENET   storage            0      enabled  offline       30
        e5b     ENET   storage          100      enabled  online        30

node2
        e0c     ENET  storage           100      enabled  online        30
        e0d     ENET  storage             0      enabled  offline       30
        e5a     ENET  storage             0      enabled  offline       30
        e5b     ENET  storage           100      enabled  online        30
```

1. [[step12]確認HA配對1的儲存組態正確無誤、而且沒有錯誤：「System Switch Ethernet show」（系統交換器乙太網路顯示）

```
storage::*> system switch ethernet show
Switch                    Type                    Address          Model
------------------------- ----------------------- ----------------
----------
sh1
                          storage-network         172.17.227.5     C9336C

      Serial Number: FOC221206C2
       Is Monitored: true
             Reason: None
   Software Version: Cisco Nexus Operating System (NX-OS) Software,
Version
                     9.3(5)
     Version Source: CDP
sh2
                          storage-network         172.17.227.6     C9336C
      Serial Number: FOC220443LZ
       Is Monitored: true
             Reason: None
   Software Version: Cisco Nexus Operating System (NX-OS) Software,
Version
                     9.3(5)
     Version Source: CDP
2 entries were displayed.
storage::*>
```

1. [[step13]重新設定HA配對1上未使用的（控制器）二線儲存連接埠、從儲存設備到網路。如果直接連接多個NS224、則應該重新設定連接埠。

```
storage port modify -node [node name] -port [port name] -mode network
```

若要將儲存連接埠放入廣播網域：

- 「網路連接埠廣播網域建立」（如有需要、可建立新網域）
- 「網路連接埠廣播網域附加連接埠」（將連接埠新增至現有網域）
  1. [[step14]如果您禁止自動建立案例、請叫用AutoSupport 下列訊息來重新啟用：「System nodeeseInvoke AutoSupport -Node *-type all -Message MAn=end」

透過新增兩個共用交換器、從具有直接附加儲存設備的交換式組態進行移轉

使用直接附加儲存設備從交換式組態移轉

當您使用Cisco Nexus 9336C-FX2共用交換器來取代某些舊版Cisco Nexus叢集交換器時、必須注意特定的組態資訊、連接埠連線和纜線需求。

- 支援下列交換器：
  - Nexus 9336C-FX2
  - Nexus 3232C
- 交換器使用下列連接埠來連線至節點：
- Nexus 9336C-FX2：
  - 連接埠1-3：離線模式（4個10G）叢集內連接埠、int E1/1/1-4、E1/2/1-4
  - 連接埠4-6：中斷模式（4個25G）叢集內/ HA連接埠、int E1/4/1-4、E1/5/1-4
  - 連接埠7-34：40/100GbE叢集內/ HA連接埠、int E1/7-34
- Nexus 3232C：
  - 連接埠1-30：10/40/100 GbE
- 交換器使用下列交換器間連結（ISL）連接埠：
  - 連接埠int E1/35-36：Nexus 9336C-FX2
  - 連接埠E1/31至32：Nexus 3232C
- 包含所有叢集交換器所支援的纜線佈線相關資訊。Hardware Universe

  請參閱 "Hardware Universe" 以取得更多資訊。

- 您已將Nexus 9336C-FX2交換器上的部分連接埠設定為以100 GbE執行。
- 您已規劃、移轉並記錄從節點到Nexus 9336C-FX2交換器的100 GbE連線。
- 本程序所支援的支援的支援版本包括Cisco乙太網路交換器頁面。ONTAP請參閱 "Cisco乙太網路交換器"。

- 您可以在不中斷營運的情況下、將其他Cisco叢集交換器從ONTAP 一個故障叢集移轉至Cisco Nexus 9336C-FX2網路交換器。

開始之前

- 現有的交換器網路必須正確設定並正常運作。
- 所有連接埠都必須處於「* up」狀態、才能確保不中斷營運。
- Nexus 9336C-FX2交換器必須設定並在適當版本的NX-OS安裝和套用參考組態檔（RCF）下運作。
- 現有的網路組態必須具備下列項目：
  - 使用兩個舊Cisco交換器的備援且功能完整的NetApp叢集。
  - 管理連線能力、以及對舊Cisco交換器和新交換器的主控台存取。
  - 使用叢集生命區的所有叢集生命區都位於其主連接埠上。
  - 在其他Cisco交換器之間以及新交換器之間啟用或連接ISL連接埠。

本程序中的範例使用下列交換器和節點命名法：

- 現有的Cisco Nexus 3232C叢集交換器為_C1_和_C2_。
- 新的Nexus 9336C-FX2交換器為_SH1_和_sh2_。
- 節點是_node1_和_node2_。
- 叢集生命區分別是節點1上的_node1_clus1_和節點1上的_node1_clus2_、以及節點2上的節點2_node2_clus1_和節點2_clus2_。
- 交換器C2會先由交換器sh2取代、然後由交換器SH1取代交換器C1。

步驟

1. 如果AutoSupport 此叢集啟用了「支援功能」、請叫用AutoSupport 下列訊息來抑制自動建立案例：「System Node AutoSupport Rsepooke -Node *-type all -most MAn=x h」

   其中x是維護時段的持續時間（以小時為單位）。

2. 檢查每個叢集連接埠的管理和作業狀態。
3. 驗證所有叢集連接埠是否都正常運作：「network port show–role cluster」（網路連接埠顯示–角色叢集）

```
cluster1::*> network port show -role cluster
Node: node1
                                                                      Ignore
                                                  Speed(Mbps)  Health   Health
Port    IPspace   Broadcast Domain Link MTU  Admin/Ope    Status   Status
------- --------- ---------------- ---- ---- ------------ -------- ------
e3a     Cluster   Cluster          up   9000  auto/100000 healthy  false
e3b     Cluster   Cluster          up   9000  auto/100000 healthy  false

Node: node2
                                                                      Ignore
                                                  Speed(Mbps) Health    Health
Port    IPspace   Broadcast Domain Link MTU  Admin/Oper  Status    Status
------- --------- ---------------- ---- ---- ----------- --------- ------
e3a     Cluster   Cluster          up   9000  auto/100000 healthy  false
e3b     Cluster   Cluster          up   9000  auto/100000 healthy  false
4 entries were displayed.
cluster1::*>
```

1. 確認所有叢集介面（lifs）都位於主連接埠：「網路介面show -role cluster」（網路介面show -role cluster）

```
cluster1::*> network interface show -role cluster
         Logical     Status     Network           Current     Current Is
Vserver  Interface   Admin/Oper Address/Mask      Node        Port    Home
-------  ----------- ---------- ----------------- ----------- ------- ----
Cluster
        node1_clus1  up/up      169.254.3.4/23    node1       e3a     true
        node1_clus2  up/up      169.254.3.5/23    node1       e3b     true
        node2_clus1  up/up      169.254.3.8/23    node2       e3a     true
        node2_clus2  up/up      169.254.3.9/23    node2       e3b     true
4 entries were displayed.
cluster1::*>
```

1. [[step5]驗證叢集是否顯示兩個叢集交換器的資訊：「系統叢集交換器show -is監控啟用運作的true」

```
cluster1::*> system cluster-switch show -is-monitoring-enabled-operational
true
Switch                    Type               Address          Model
------------------------- ------------------ ---------------- ------
sh1                       cluster-network    10.233.205.90    N9K-C9336C
     Serial Number: FOCXXXXXXGD
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
    Version Source: CDP
sh2                       cluster-network    10.233.205.91    N9K-C9336C
     Serial Number: FOCXXXXXXGS
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
    Version Source: CDP
cluster1::*>
```

1. 停用叢集生命體上的自動還原功能。

```
cluster1::*> network interface modify -vserver Cluster -lif * -auto-revert
false
```

1. [[step7]關閉C2交換器：

```
c2# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
c2(config)# interface ethernet <int range>
c2(config)#shutdown
```

1. [[step8]請確認叢集lifs已移轉至叢集交換器SH1上裝載的連接埠：「網路介面show -role cluster」（網路介面顯示角色叢集）、這可能需要幾秒鐘的時間。

```
cluster1::*> network interface show -role cluster
          Logical     Status     Network         Current      Current  Is
Vserver   Interface   Admin/Oper Address/Mask    Node         Port
Home
--------- ----------- ---------- --------------- ------------ --------
-----
Cluster
          node1_clus1 up/up      169.254.3.4/23  node1        e3a
true
          node1_clus2 up/up      169.254.3.5/23  node1        e3a
false
          node2_clus1 up/up      169.254.3.8/23  node2        e3a
true
          node2_clus2 up/up      169.254.3.9/23  node2        e3a
false
4 entries were displayed.
cluster1::*>
```

1. 將交換器C2更換為新交換器sh2、然後重新連接新交換器。
2. 確認sh2上的連接埠已備份。*注意*生命仍在交換器C1上。
3. 關閉C1交換器：

```
c1# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
c1(config)# interface ethernet <int range>
c1(config)#shutdown
```

1. [[step12]驗證叢集LIF是否已移轉至叢集交換器sh2上裝載的連接埠。這可能需要幾秒鐘的時間。

```
cluster1::*> network interface show -role cluster
         Logical        Status     Network         Current   Current Is
Vserver  Interface      Admin/Oper Address/Mask    Node      Port    Home
-------- -------------- ---------- --------------- --------- ------- ----
Cluster
         node1_clus1    up/up      169.254.3.4/23  node1     e3a     true
         node1_clus2    up/up      169.254.3.5/23  node1     e3a     false
         node2_clus1    up/up      169.254.3.8/23  node2     e3a     true
         node2_clus2    up/up      169.254.3.9/23  node2     e3a     false
4 entries were displayed.
cluster1::*>
```

1. [[step13]將交換器C1更換為新的交換器SH1、然後重新連接新的交換器。

2. 確認SH1上的連接埠已備份。*注意*生命仍在交換器C2上。
3. 在叢集生命體上啟用自動還原：

```
cluster1::*> network interface modify -vserver Cluster -lif * -auto-revert
True
```

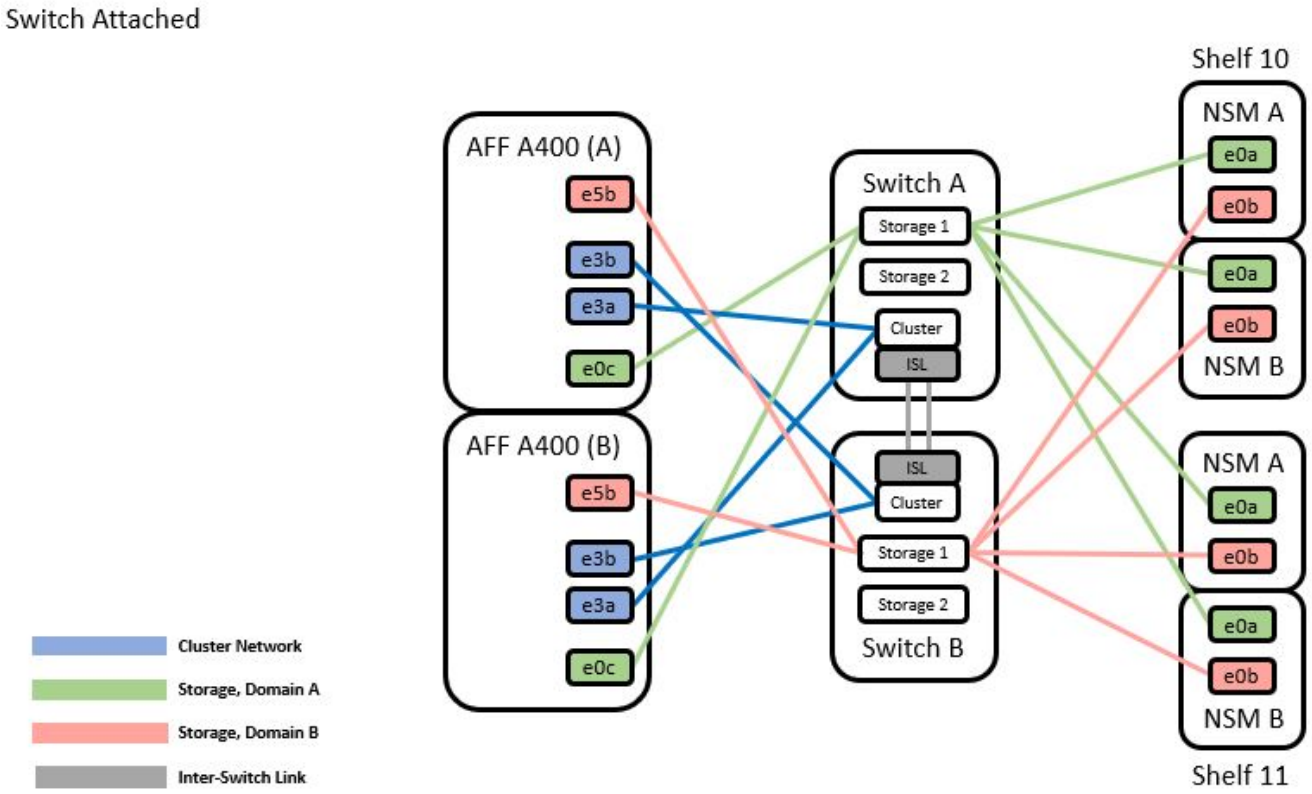
1. [[step16]驗證叢集是否正常：「叢集顯示」

```
cluster1::*> cluster show
Node                  Health  Eligibility   Epsilon
--------------------- ------- ------------- -------
node1                 true    true          false
node2                 true    true          false
2 entries were displayed.
cluster1::*>
```

重複使用儲存交換器、從交換器附加儲存設備的無交換器組態移轉

移轉儲存交換器

透過重複使用儲存交換器、HA配對1的儲存交換器便成為共享交換器。*交換器附加的佈線圖
*



步驟

1. 確認HA配對1（和HA配對2）的儲存組態正確無誤、而且沒有錯誤：「System Switch Ethernet show」（系統交換器乙太網路顯示）

```
storage::*> system switch ethernet show
Switch                    Type                  Address          Model
------------------------- --------------------- ----------------
----------
sh1
                          storage-network       172.17.227.5     C9336C

    Serial Number: FOC221206C2
     Is Monitored: true
           Reason: none
 Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                   9.3(5)
   Version Source: CDP
sh2
                        storage-network        172.17.227.6      C9336C
    Serial Number: FOC220443LZ
     Is Monitored: true
           Reason: None
 Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                   9.3(5)
   Version Source: CDP
2 entries were displayed.
storage::*>
```

1. [[step2]確認節點連接埠正常運作：「torage port show -port-type ENET」

```
storage::*> storage port show -port-type ENET
                                     Speed                            VLAN
Node     Port     Type     Mode        (Gb/s)     State      Status       ID
------- ------- ------- ---------- --------- --------- --------- -----
node1
        e0c      ENET     storage          100 enabled   online       30
        e0d      ENET     storage          100 enabled   online       30
        e5a      ENET     storage          100 enabled   online       30
        e5b      ENET     storage          100 enabled   online       30

node2
        e0c      ENET     storage          100 enabled   online       30
        e0d      ENET     storage          100 enabled   online       30
        e5a      ENET     storage          100 enabled   online       30
        e5b      ENET     storage          100 enabled   online       30
```

1. [[step3]將HA配對1、NSM224路徑A纜線從儲存交換器A移至儲存交換器A上的HA配對1、路徑A的共享NS224儲存連接埠
2. 將纜線從HA配對1、節點A、路徑A移至HA配對1、儲存交換器A上節點A的共用儲存連接埠
3. 將纜線從HA配對1、節點B、路徑A移至儲存交換器A上的HA配對1、節點B共用儲存連接埠
4. 驗證連接至HA配對1的儲存設備、儲存交換器A是否正常：「系統健全狀況警示顯示-instance」

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step7]將共享交換器A上的儲存RCF替換為共用RCF檔案。請參閱 "在Cisco Nexus 9336C-FX2共享交換器上安裝RCF" 以取得更多詳細資料。
2. 驗證連接至HA配對1的儲存設備、儲存交換器B是否正常：「系統健全狀況警示顯示-instance」

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step9]將HA配對1、NSM224路徑B纜線從儲存交換器B移至HA配對1的共享NS224儲存連接埠、將路徑B移至儲存交換器B
2. 將纜線從HA配對1、節點A、路徑B移至HA配對1、節點A、儲存交換器B上路徑B的共享儲存連接埠
3. 將纜線從HA配對1、節點B、路徑B移至HA配對1、節點B、儲存交換器B路徑B的共享儲存連接埠
4. 驗證連接至HA配對1的儲存設備、儲存交換器B是否正常：「系統健全狀況警示顯示-instance」

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step13]將共享交換器B上的儲存RCF檔案替換為共用RCF檔案。請參閱 "在Cisco Nexus 9336C-FX2共享交換器上安裝RCF" 以取得更多詳細資料。
2. 驗證連接至HA配對1的儲存設備、儲存交換器B是否正常：「系統健全狀況警示顯示-instance」

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step15]在共享交換器A和共享交換器B之間安裝ISL：

```
sh1# configure
Enter configuration commands, one per line. End with CNTL/Z.
sh1 (config)# interface e1/35-36
sh1 (config-if-range)# no lldp transmit
sh1 (config-if-range)# no lldp receive
sh1 (config-if-range)# switchport mode trunk
sh1 (config-if-range)# no spanning-tree bpduguard enable
sh1 (config-if-range)# channel-group 101 mode active
sh1 (config-if-range)# exit
sh1 (config)# interface port-channel 101
sh1 (config-if)# switchport mode trunk
sh1 (config-if)# spanning-tree port type network
sh1 (config-if)# exit
sh1 (config)# exit
```

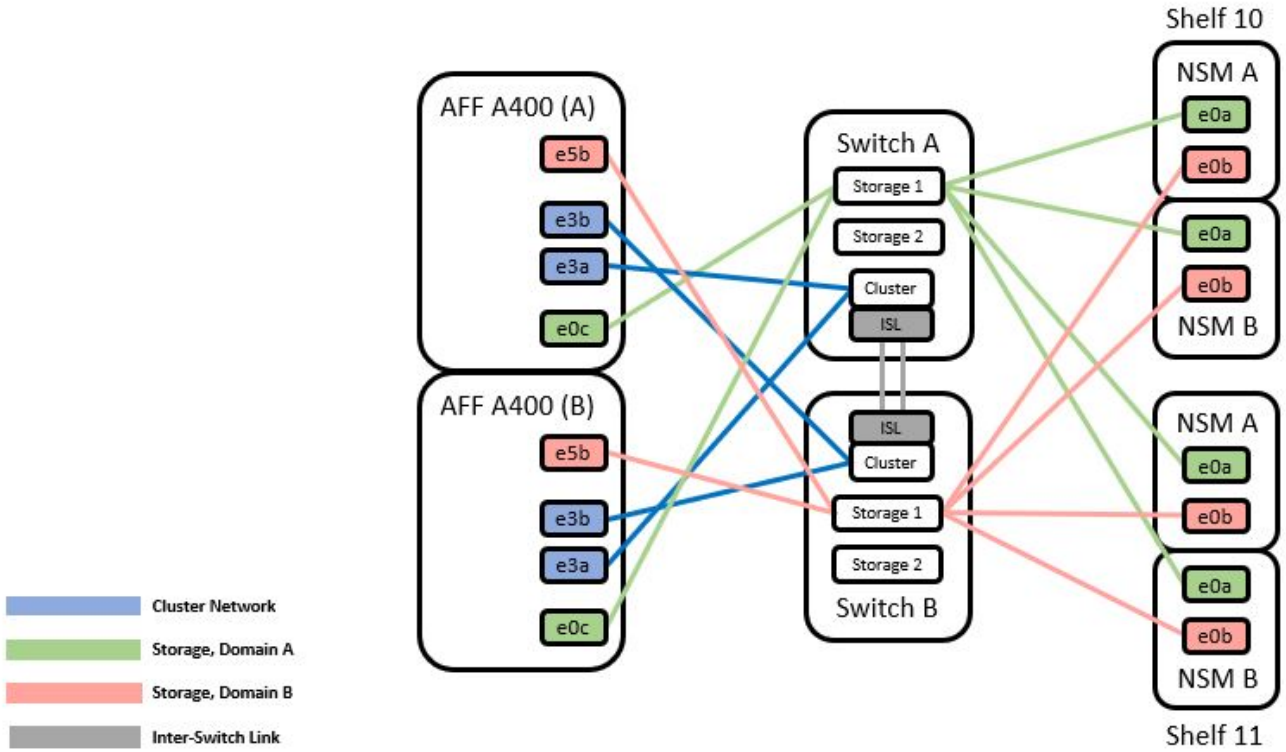
1. [[step16]將HA配對1從無交換式叢集轉換成交換式叢集。使用共享RCF定義的叢集連接埠指派。請參閱 "安裝NX-OS軟體與參考組態檔（RCT）"以取得更多詳細資料。
2. 驗證交換式網路組態是否有效：「network port show」（網路連接埠顯示）

重複使用儲存交換器、從交換式叢集與交換式附加儲存設備進行移轉

移轉儲存交換器

透過重複使用儲存交換器、HA配對1的儲存交換器便成為共享交換器。*交換器附加的佈線圖
*

Switch Attached



步驟

1. 確認HA配對1（和HA配對2）的儲存組態正確無誤、而且沒有錯誤：「System Switch Ethernet show」（系統交換器乙太網路顯示）

```
storage::*> system switch ethernet show
Switch                    Type               Address          Model
------------------------- ------------------ ---------------- ----------
sh1
                          storage-network    172.17.227.5     C9336C

     Serial Number: FOC221206C2
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
       Version Source: CDP
sh2
                          storage-network    172.17.227.6     C9336C
     Serial Number: FOC220443LZ
      Is Monitored: true
            Reason: None
  Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                    9.3(5)
    Version Source: CDP
2 entries were displayed.
storage::*>
```

1. 將HA配對1、NSM224路徑A纜線從儲存交換器A移至儲存交換器A上的HA配對1、路徑A的NSM224儲存連接埠
2. 將纜線從HA配對1、節點A、路徑A移至儲存交換器A上的NSM224儲存連接埠
3. 將纜線從HA配對1、節點B、路徑A移至儲存交換器A上的NSM224儲存連接埠、以供HA配對1使用
4. 驗證連接至HA配對1的儲存設備、儲存交換器A是否正常：「torage port show -port-ttype ENET」

```
storage::*> storage port show -port-type ENET
                                     Speed                              VLAN
Node     Port     Type     Mode        (Gb/s)        State     Status       ID
------- ------- ------- ---------- ------------ --------- --------- -----
node1
        e0c      ENET     storage            100  enabled   online       30
        e0d      ENET     storage            100  enabled   online       30
        e5a      ENET     storage            100  enabled   online       30
        e5b      ENET     storage            100  enabled   online       30

node2
        e0c      ENET     storage            100  enabled   online       30
        e0d      ENET     storage            100  enabled   online       30
        e5a      ENET     storage            100  enabled   online       30
        e5b      ENET     storage            100  enabled   online       30
```

1. [[step6]將共享交換器A上的儲存RCF替換為共用RCF檔案。請參閱 "在Cisco Nexus 9336C-FX2共享交換器上安裝RCF" 以取得更多詳細資料。
2. 驗證連接至HA配對1的儲存設備、儲存交換器A是否正常：「系統健全狀況警示顯示-instance」

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step8]將HA配對1、NSM224路徑B纜線從儲存交換器B移至HA配對1的共享NS224儲存連接埠、將路徑B移至儲存交換器B
2. 將纜線從HA配對1、節點A、路徑B移至HA配對1、節點A、儲存交換器B上路徑B的共享儲存連接埠
3. 將纜線從HA配對1、節點B、路徑B移至HA配對1、節點B、儲存交換器B路徑B的共享儲存連接埠
4. 驗證連接至HA配對1的儲存設備、儲存交換器B是否正常：「系統健全狀況警示顯示-instance」

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step12]將共享交換器B上的儲存RCF檔案替換為共用RCF檔案。請參閱 "在Cisco Nexus 9336C-FX2共享交換器上安裝RCF" 以取得更多詳細資料。
2. 驗證連接至HA配對1的儲存設備、儲存交換器B是否正常：「系統健全狀況警示顯示-instance」

```
storage::*> system health alert show -instance
There are no entries matching your query.
```

1. [[step14]確認HA配對1的儲存組態正確無誤、而且沒有錯誤：「System Switch Ethernet show」（系統交換器乙太網路顯示）

```
storage::*> system switch ethernet show
Switch                      Type                 Address          Model
--------------------------- -------------------- ---------------- ----------
sh1
                            storage-network      172.17.227.5     C9336C

    Serial Number: FOC221206C2
     Is Monitored: true
           Reason: None
 Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                   9.3(5)
   Version Source: CDP
sh2
                            storage-network      172.17.227.6     C9336C
    Serial Number: FOC220443LZ
     Is Monitored: true
           Reason: None
 Software Version: Cisco Nexus Operating System (NX-OS) Software, Version
                   9.3(5)
   Version Source: CDP
2 entries were displayed.
storage::*>
```

1. [[step15]在共享交換器A和共享交換器B之間安裝ISL：

```
sh1# configure
Enter configuration commands, one per line. End with CNTL/Z.
sh1 (config)# interface e1/35-36
sh1 (config-if-range)# no lldp transmit
sh1 (config-if-range)# no lldp receive
sh1 (config-if-range)# switchport mode trunk
sh1 (config-if-range)# no spanning-tree bpduguard enable
sh1 (config-if-range)# channel-group 101 mode active
sh1 (config-if-range)# exit
sh1 (config)# interface port-channel 101
sh1 (config-if)# switchport mode trunk
sh1 (config-if)# spanning-tree port type network
sh1 (config-if)# exit
sh1 (config)# exit
```

1. [[step16]使用交換器更換程序和共用RCF、將叢集網路從現有的叢集交換器移轉到共用交換器。新的共享交換器A是「CS1」。新的共享交換器B是「CS2」。請參閱 "更換Cisco Nexus 9336C-FX2共享交換器" 和 "在Cisco Nexus 9336C-FX2共享交換器上安裝RCF" 以取得更多詳細資料。
2. 驗證交換式網路組態是否有效：「network port show」（網路連接埠顯示）

3. 移除未使用的叢集交換器。
4. 移除未使用的儲存交換器。

更換Cisco Nexus 9336C-FX2共享交換器

更換Cisco Nexus 9336C-FX2共享交換器

更換故障的Nexus 9336C-FX2共用交換器是不中斷營運的程序（NDU）。

在目前環境和更換交換器上執行交換器更換之前、必須符合下列條件。

- 現有的叢集與網路基礎架構：
  - 必須使用至少一個完全連接的叢集交換器、驗證現有叢集是否完全正常運作。
  - 所有叢集連接埠必須為* up *。
  - 所有叢集邏輯介面（LIF）都必須為* up *及其主連接埠。
  - 使用支援Ping叢ONTAP 集的節點節點node1命令時、必須指出所有路徑上的基本連線能力和大於PMTU的通訊均已成功完成。
- Nexus 9336C-FX2替換交換器：
  - 替換交換器上的管理網路連線功能必須正常運作。
  - 更換交換器的主控台存取必須已就緒。
  - 節點連線為連接埠1/1至1/34：
  - 所有交換器間連結（ISL）連接埠必須在連接埠1/35和1/36上停用。
  - 所需的參考組態檔（RCF）和NX-OS作業系統映像交換器必須載入交換器。
  - 任何先前的站台自訂（例如STP、SNMP和SSH）都應該複製到新交換器。

您必須執行命令、從裝載叢集LIF的節點移轉叢集LIF。

本程序中的範例使用下列交換器和節點命名法：

- 現有Nexus 9336C-FX2交換器的名稱為_SH1_和_sh2_。
- 新Nexus 9336C-FX2交換器的名稱為_newsh1_和_newsh2_。
- 節點名稱是_node1_和_node2_。
- 每個節點上的叢集連接埠名稱為_e3a_和_e3c_。
- 叢集LIF名稱為節點1的「node1_clus1」和「node1_clus2」、節點2的「node2_clus1」和「node2_clus2」。
- 對所有叢集節點進行變更的提示為cluster1：：*>。

> 下列程序以下列網路拓撲為基礎：

```
cluster1::*> network port show -ipspace Cluster
```

```
Node: node1

Ignore
                                                      Speed(Mbps)  Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
Status
--------- ------------ ---------------- ---- ---- ------------ --------
------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
false
e3b       Cluster      Cluster          up   9000  auto/100000 healthy
false

Node: node2

Ignore
                                                      Speed(Mbps)  Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
Status
--------- ------------ ---------------- ---- ---- ------------ --------
------
e3a       Cluster      Cluster          up   9000  auto/100000 healthy
false
e3b       Cluster      Cluster          up   9000  auto/100000 healthy
false
4 entries were displayed.


cluster1::*> network interface show -vserver Cluster
            Logical    Status     Network            Current       Current
Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ ------------- -------
----
Cluster
            node1_clus1  up/up    169.254.209.69/16  node1         e3a
true
            node1_clus2  up/up    169.254.49.125/16  node1         e3b
true
            node2_clus1  up/up    169.254.47.194/16  node2         e3a
true
            node2_clus2  up/up    169.254.19.183/16  node2         e3b
true
```

```
4 entries were displayed.

cluster1::*> network device-discovery show -protocol cdp
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------
----------------
node2      /cdp
            e3a    sh1                       Eth1/2            N9K-C9336C
            e3b    sh2                       Eth1/2            N9K-C9336C

node1      /cdp
            e3a    sh1                       Eth1/1            N9K-C9336C
            e3b    sh2                       Eth1/1            N9K-C9336C
4 entries were displayed.

sh1# show cdp neighbors
Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute
Device-ID          Local Intrfce  Hldtme Capability  Platform      Port ID
node1              Eth1/1         144    H           FAS2980       e3a
node2              Eth1/2         145    H           FAS2980       e3a
sh2                Eth1/35        176    R S I s     N9K-C9336C    Eth1/35
sh2 (FDO220329V5)   Eth1/36       176    R S I s     N9K-C9336C    Eth1/36
Total entries displayed: 4

sh2# show cdp neighbors
Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute
Device-ID          Local Intrfce  Hldtme Capability  Platform      Port ID
node1              Eth1/1         139    H           FAS2980       eb
node2              Eth1/2         124    H           FAS2980       eb
sh1                Eth1/35        178    R S I s     N9K-C9336C    Eth1/35
sh1                Eth1/36        178    R S I s     N9K-C9336C    Eth1/36
Total entries displayed: 4
```

步驟

1. 如果AutoSupport 此叢集啟用了「支援功能」、請叫用AutoSupport 「系統節點AutoSupport 支援功能叫用節點*-type all -most MAn=xh」訊息來禁止自動建立案例

   其中x是維護時段的持續時間（以小時為單位）。

2. 選用：在交換器newsh2上安裝適當的RCF和映像、並進行必要的站台準備。
   a. 如有必要、請針對新交換器驗證、下載及安裝適當版本的RCF和NX-OS軟體。如果您已確認新交換器已正確設定、而且不需要更新RCF和NX-OS軟體、請繼續執行 步驟3.。
   b. 請前往NetApp支援網站上的NetApp叢集與管理網路交換器參考組態檔案說明頁面。
   c. 按一下叢集網路與管理網路相容性對照表的連結、然後記下所需的交換器軟體版本。
   d. 按一下瀏覽器的返回箭號以返回「Description（說明）」頁面、按一下「Continue（繼續）」、「Accept the license agreement（接受授權合約）」、然後前往「Download（下載）」頁面。
   e. 請依照下載頁面上的步驟、針對ONTAP 您要安裝的版本的更新軟體、下載正確的RCF和NX-OS檔案。
3. [[step3]在新交換器上、以admin身分登入、並關閉所有連接至節點叢集介面的連接埠（連接埠1/1至1/34）。如果您要更換的交換器無法正常運作且已關機、請前往 步驟4.。叢集節點上的LIF應已容錯移轉至每個節點的其他叢集連接埠。

```
newsh2# config
Enter configuration commands, one per line. End with CNTL/Z.
newsh2(config)# interface e1/1-34
newsh2(config-if-range)# shutdown
```

1. 確認所有叢集生命週年都已啟用自動還原功能。「網路介面show - vserver叢集-功能變數自動回復」

```
cluster1::> network interface show -vserver Cluster -fields auto-revert
             Logical
Vserver      Interface     Auto-revert
------------ ------------- -------------
Cluster      node1_clus1   true
Cluster      node1_clus2   true
Cluster      node2_clus1   true
Cluster      node2_clus2   true
4 entries were displayed.
```

1. [[step5]驗證所有叢集l生命 週年是否都能通訊：「叢集ping cluster <節點名稱>」

```
cluster1::*> cluster ping-cluster node1
Host is node2
Getting addresses from network interface table...
Cluster node1_clus1 169.254.209.69 node1 e3a
Cluster node1_clus2 169.254.49.125 node1 e3b
Cluster node2_clus1 169.254.47.194 node2 e3a
Cluster node2_clus2 169.254.19.183 node2 e3b
Local = 169.254.47.194 169.254.19.183
Remote = 169.254.209.69 169.254.49.125
Cluster Vserver Id = 4294967293
Ping status:
....
Basic connectivity succeeds on 4 path(s)
Basic connectivity fails on 0 path(s)
................
Detected 9000 byte MTU on 4 path(s):
Local 169.254.47.194 to Remote 169.254.209.69
Local 169.254.47.194 to Remote 169.254.49.125
Local 169.254.19.183 to Remote 169.254.209.69
Local 169.254.19.183 to Remote 169.254.49.125
Larger than PMTU communication succeeds on 4 path(s)
RPC status:
2 paths up, 0 paths down (tcp check)
2 paths up, 0 paths down (udp check)
```

1. [[step6]關閉Nexus 9336C-FX2交換器SH1上的ISL連接埠1/35和1/36。

```
sh1# configure
Enter configuration commands, one per line. End with CNTL/Z.
sh1(config)# interface e1/35-36
sh1(config-if-range)# shutdown
sh1(config-if-range)#
```

1. [[step7]從Nexus 9336C-FX2 sh2交換器拔下所有纜線、然後將其連接至Nexus C9336C-FX2 newsh2交換器上的相同連接埠。
2. 在SH1和newsh2交換器之間啟動ISL連接埠1/35和1/36、然後驗證連接埠通道作業狀態。

   連接埠通道應指示PO1（SU）、而成員連接埠應指示eth1/35（P）和eth1/36（P）。

   此範例可啟用ISL連接埠1/35和1/36、並在交換器SH1上顯示連接埠通道摘要。

```
sh1# configure
Enter configuration commands, one per line. End with CNTL/Z.
sh1 (config)# int e1/35-36
sh1 (config-if-range)# no shutdown
sh1 (config-if-range)# show port-channel summary
Flags:  D - Down        P - Up in port-channel (members)
        I - Individual  H - Hot-standby (LACP only)
        s - Suspended   r - Module-removed
        b - BFD Session Wait
        S - Switched    R - Routed
        U - Up (port-channel)
        p - Up in delay-lacp mode (member)
        M - Not in use. Min-links not met
--------------------------------------------------------------------------
------
Group Port-       Type     Protocol  Member       Ports
      Channel
--------------------------------------------------------------------------
------
1     Po1(SU)     Eth      LACP      Eth1/35(P)   Eth1/36(P)

sh1 (config-if-range)#
```

1. 確認所有節點上的連接埠e3b都已開啟：「網路連接埠顯示IPSpace叢集」

   輸出應如下所示：

```
cluster1::*> network port show -ipspace Cluster

Node: node1

Ignore
                                                      Speed(Mbps)  Health
Health
Port      IPspace      Broadcast Domain Link MTU   Admin/Oper   Status
Status
--------- ------------ ---------------- ---- ----- ---------- - - --------
----
e3a       Cluster      Cluster          up   9000  auto/100000  healthy
false
e3b       Cluster      Cluster          up   9000  auto/100000  healthy
false

Node: node2

Ignore
                                                      Speed(Mbps)  Health
Health
Port      IPspace      Broadcast Domain Link MTU   Admin/Oper   Status
Status
--------- ------------ ---------------- ---- ----- ----------- -  --------
----
e3a       Cluster      Cluster          up   9000  auto/100000  healthy
false
e3b       Cluster      Cluster          up   9000  auto/auto    -
false
4 entries were displayed.
```

1. [[step10]在上一步所用的同一個節點上、使用network interface revert命令、還原上一步與連接埠相關聯的叢集LIF。

   在此範例中、如果Home值為true且連接埠為e3b、則節點1上的LIF node1_clus2會成功還原。

   下列命令會將節點1上的LIF node1_clus2傳回主連接埠e3a、並顯示兩個節點上的LIF LIF相關資訊。如果兩個叢集介面的「Is Home」欄位都為* true*、而且顯示正確的連接埠指派、則啟動第一個節點會成功、在此範例中、節點1上的「是Home」欄位和「e3b」。

```
cluster1::*> network interface show -vserver Cluster

            Logical      Status     Network            Current    Current
Is
Vserver     Interface    Admin/Oper Address/Mask       Node       Port
Home
----------- ------------ ---------- ------------------ ---------- -------
-----
Cluster
            node1_clus1  up/up      169.254.209.69/16  node1      e3a
true
            node1_clus2  up/up      169.254.49.125/16  node1      e3b
true
            node2_clus1  up/up      169.254.47.194/16  node2      e3a
true
            node2_clus2  up/up      169.254.19.183/16  node2      e3a
false
4 entries were displayed.
```

1. [[step11]顯示叢集中節點的相關資訊：「叢集顯示」

   此範例顯示此叢集中節點1和節點2的節點健全狀況為真：

```
cluster1::*> cluster show
Node          Health  Eligibility
------------- ------- ------------
node1         false   true
node2         true    true
```

1. [[step12]確認所有實體叢集連接埠都已啟動：「網路連接埠顯示IPSpace叢集」

```
cluster1::*> network port show -ipspace Cluster

Node node1
Ignore
                                                      Speed(Mbps)  Health
Health
Port      IPspace     Broadcast Domain  Link  MTU   Admin/Oper   Status
Status
--------- ----------- ----------------- ----- ----- ------------ --------
------
e3a       Cluster     Cluster           up    9000  auto/100000  healthy
false
e3b       Cluster     Cluster           up    9000  auto/100000  healthy
false

Node: node2

Ignore
                                                      Speed(Mbps)  Health
Health
Port      IPspace      Broadcast Domain Link  MTU   Admin/Oper   Status
Status
--------- ------------ ---------------- ----- ----- ------------ --------
------
e3a       Cluster      Cluster          up    9000  auto/100000  healthy
false
e3b       Cluster      Cluster          up    9000  auto/100000  healthy
false
4 entries were displayed.
```

1. [[step13]驗證所有叢集生命體是否都能通訊：「叢集ping叢集」

```
cluster1::*> cluster ping-cluster -node node2
Host is node2
Getting addresses from network interface table...
Cluster node1_clus1 169.254.209.69 node1 e3a
Cluster node1_clus2 169.254.49.125 node1 e3b
Cluster node2_clus1 169.254.47.194 node2 e3a
Cluster node2_clus2 169.254.19.183 node2 e3b
Local = 169.254.47.194 169.254.19.183
Remote = 169.254.209.69 169.254.49.125
Cluster Vserver Id = 4294967293
Ping status:
....
Basic connectivity succeeds on 4 path(s)
Basic connectivity fails on 0 path(s)
................
Detected 9000 byte MTU on 4 path(s):
Local 169.254.47.194 to Remote 169.254.209.69
Local 169.254.47.194 to Remote 169.254.49.125
Local 169.254.19.183 to Remote 169.254.209.69
Local 169.254.19.183 to Remote 169.254.49.125
Larger than PMTU communication succeeds on 4 path(s)
RPC status:
2 paths up, 0 paths down (tcp check)
2 paths up, 0 paths down (udp check)
```

1. [[step14]確認下列叢集網路組態：「network port show」

```
cluster1::*> network port show -ipspace Cluster

Node: node1

Ignore
                                          Speed(Mbps)             Health
Health
Port      IPspace     Broadcast Domain Link MTU   Admin/Oper   Status
Status
--------- ----------- ---------------- ---- ----- ------------ --------
------
e3a       Cluster     Cluster          up   9000  auto/100000  healthy
false
e3b       Cluster     Cluster          up   9000  auto/100000  healthy
false

Node: node2
```

```
Ignore
                                          Speed(Mbps)            Health
Health
Port      IPspace      Broadcast Domain Link MTU  Admin/Oper   Status
Status
--------- ------------ ---------------- ---- ---- ------------ --------
------
e3a       Cluster      Cluster          up   9000 auto/100000  healthy
false
e3b       Cluster      Cluster          up   9000 auto/100000  healthy
false
4 entries were displayed.

cluster1::*> network interface show -vserver Cluster
            Logical    Status     Network            Current       Current
Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ ------------- -------
----
Cluster
            node1_clus1  up/up    169.254.209.69/16  node1         e3a
true
            node1_clus2  up/up    169.254.49.125/16  node1         e3b
true
            node2_clus1  up/up    169.254.47.194/16  node2         e3a
true
            node2_clus2  up/up    169.254.19.183/16  node2         e3b
true
4 entries were displayed.

cluster1::> network device-discovery show -protocol cdp
Node/       Local  Discovered
Protocol    Port   Device (LLDP: ChassisID)  Interface         Platform
----------- ------ ------------------------- ----------------
----------------
node2      /cdp
            e3a    sh1    0/2               N9K-C9336C
            e3b    newsh2                    0/2               N9K-C9336C
node1      /cdp
            e3a    sh1                       0/1               N9K-C9336C
            e3b    newsh2                    0/1               N9K-C9336C
4 entries were displayed.

sh1# show cdp neighbors
```

```
Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute
Device-ID            Local Intrfce  Hldtme Capability  Platform      Port
ID
node1                Eth1/1         144    H           FAS2980       e3a
node2                Eth1/2         145    H           FAS2980       e3a
newsh2               Eth1/35        176    R S I s     N9K-C9336C
Eth1/35
newsh2               Eth1/36        176    R S I s     N9K-C9336C
Eth1/36
Total entries displayed: 4

sh2# show cdp neighbors
Capability Codes: R - Router, T - Trans-Bridge, B - Source-Route-Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  V - VoIP-Phone, D - Remotely-Managed-Device,
                  s - Supports-STP-Dispute
Device-ID          Local Intrfce  Hldtme Capability  Platform      Port ID
node1              Eth1/1         139    H           FAS2980       e3b
node2              Eth1/2         124    H           FAS2980       eb
sh1                Eth1/35        178    R S I s     N9K-C9336C    Eth1/35
sh1                Eth1/36        178    R S I s     N9K-C9336C    Eth1/36
Total entries displayed: 4
```

1. [[step15]使用下列命令、啟用乙太網路交換器健全狀況監視器記錄收集功能、以收集交換器相關的記錄檔：
   - '系統交換器乙太網路記錄設定密碼'
   - 「系統交換器乙太網路記錄啟用收集」

```
cluster1::*> system switch ethernet log setup-password
Enter the switch name: <return>
The switch name entered is not recognized.
Choose from the following list:
sh1
sh2
cluster1::*> system switch ethernet log setup-password
Enter the switch name: sh1
RSA key fingerprint is e5:8b:c6:dc:e2:18:18:09:36:63:d9:63:dd:03:d9:cc
Do you want to continue? {y|n}::[n] y
Enter the password: <enter switch password>
Enter the password again: <enter switch password>
cluster1::*> system switch ethernet log setup-password
Enter the switch name: sh2
RSA key fingerprint is 57:49:86:a1:b9:80:6a:61:9a:86:8e:3c:e3:b7:1f:b1
Do you want to continue? {y|n}:: [n] y
Enter the password: <enter switch password>
Enter the password again: <enter switch password>
cluster1::*> system  switch ethernet log enable-collection
Do you want to enable cluster log collection for all nodes in the cluster?
y|n}: [n] y
Enabling cluster switch log collection.
cluster1::*>
```

如果這些命令中有任何一個出現錯誤、請聯絡NetApp支援部門。

1. [[step16]將儲存連接埠從舊交換器sh2移至新交換器newsh2。
2. 驗證連接至HA配對1的儲存設備、共享交換器newsh2是否正常。
3. 驗證連接至HA配對2的儲存設備、共享交換器newsh2是否正常：「torage port show -port-type ENET」

```
storage::*> storage port show -port-type ENET
                                     Speed                              VLAN
Node     Port     Type     Mode       (Gb/s)       State      Status       ID
------- ------- ------- ---------- ----------- --------- --------- -----
node1
        e3a      ENET     storage          100   enabled    online       30
        e3b      ENET     storage            0   enabled    offline      30
        e7a      ENET     storage            0   enabled    offline      30
        e7b      ENET     storage          100   enabled    online       30

node2
        e3a      ENET     storage          100   enabled    online       30
        e3b      ENET     storage            0   enabled    offline      30
        e7a      ENET     storage            0   enabled    offline      30
        e7b      ENET     storage          100   enabled    online       30
```

1. [[step19]驗證磁碟櫃是否已正確連接纜線：「torage shelage port show -Fields remie-device-remie-port」（磁碟櫃連接埠顯示-欄位遠端裝置、遠端連接埠）

```
cluster1::*> storage shelf port show -fields remote-device,remote-port
shelf id remote-port  remote-device
----- -- ------------ ----------------------------
3.20  0  Ethernet1/13 sh1
3.20  1  Ethernet1/13 newsh2
3.20  2  Ethernet1/14 sh1
3.20  3  Ethernet1/14 newsh2
3.30  0  Ethernet1/15 sh1
3.30  1  Ethernet1/15 newsh2
3.30  2  Ethernet1/16 sh1
3.30  3  Ethernet1/16 newsh2
8 entries were displayed.
```

1. [[step20]移除舊交換器sh2。
2. 對交換器SH1和新交換器newsh1重複這些步驟。
3. 如果您禁止自動建立個案、請叫用AutoSupport 下列訊息重新啟用此功能：「System Node AutoSupport 現象叫用節點*-type all -most MAn=end」

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